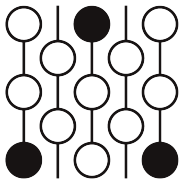
Review of Problem 1 – ProofCouncil



First Proof Third Batch
Reviewer 1
October 8–9, 2026

1 Review

This solution was clearly the best of those we reviewed. It followed the same general lines as the human solution. This solution made use of more abstract tools than the other solutions (it introduced an auxiliary topology), which made it conceptually simpler. This solution also used a Mayer–Vietoris argument where the authors had constructed a contractible simplicial complex, which gave an additional conceptual simplification. The writing is excellent overall, although there is room to tailor the presentation better for a human reader. I did not identify any inadequate citations.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score: 1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 1

2 Submission with inline comments

Abstract

Let P be an integral, saturated, sharp commutative monoid, let A_P be the category of monoids under P, equipped with the birhizomic topology on A_P^{op}, and let $B_P \subset A_P$ be the full subcategory of the sharpened localizations $P \to P[-x]^\sharp$, $x \in P$, with the trivial topology (all definitions are recalled in §1). The inclusion induces an inverse image functor π^{-1} from presheaves on B_P^{op} to birhizomic sheaves, with right adjoint π_*, and the question answered here asks whether $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism for every presheaf of abelian groups F on B_P^{op}. We show that it is, for every P, with no finiteness hypothesis. The proof replaces the birhizomic topology by a finer auxiliary topology generated by binary "sign covers" $\{R \to R\langle x\rangle,\ R \to R\langle -x\rangle\}$, where $R\langle \pm x\rangle$ is the universal monoid under R in which $\pm x \in R^{\mathrm{gp}}$ becomes nonnegative; a compactness argument shows that every birhizomic cover is a cover for it, and the left Kan extension of F turns out to be a sheaf for it with split exact Mayer–Vietoris sequences, hence acyclic by an induction on the degree. Acyclicity then descends to the coarser birhizomic topology.

3 Introduction

3.1 Conventions and the objects in the question

Throughout, a *monoid* means a commutative unital monoid M, written additively, that is *integral* (cancellative, so that $M \to M^{\mathrm{gp}}$ is injective, where M^{gp} is the group completion), *saturated* (if $g \in M^{\mathrm{gp}}$ and $ng \in M$ for some integer $n \geq 1$, then $g \in M$), and *sharp* (the only unit is 0). All monoids, homomorphisms, pushouts and colimits below are taken in this category, which we denote by **Mon**. It is a reflective subcategory of commutative monoids (the reflection integralizes, saturates and sharpens), so it has all colimits; the pushouts we actually need to compute are given explicitly in (4)–(5). Basic facts about integral and saturated monoids, localization and sharpening are collected in Ogus's book [4, Ch. I, §§1.3–1.4]; the short verifications we need are included.

Two operations on submonoids of an abelian group G recur constantly. For a submonoid $C \subset G$, the *saturation* of C in G is $C^{\mathrm{sat}} = \{g \in G : ng \in C \text{ for some } n \geq 1\}$, and the *sharpening* of C is $C^\sharp = C/C^\times$, the quotient by the group of units $C^\times = C \cap (-C)$. If C is saturated in G then $C^\sharp \subset G/C^\times$ is integral, saturated and sharp (if $n(g + C^\times) \in C/C^\times$, then $ng + h \in C$ for some $h \in C^\times$, hence $ng \in C$ and $g \in C$).

A monoid M orders its group completion: for $x, y \in M^{\mathrm{gp}}$ put $x \leq y$ if $y - x \in M$. This is a translation-invariant partial order, and M is called *valuative* if the order is total, i.e. $M \cup (-M) = M^{\mathrm{gp}}$. We note once and for all that M^{gp} is torsion-free for every monoid M: if $ng = 0$ with $n \geq 1$, saturation puts g and $-g$ in M, and sharpness forces $g = 0$.

Fix a monoid P; it is arbitrary (in particular, no finite generation is assumed), and it stays fixed throughout. Let A_P be the category of monoids under P, i.e. of homomorphisms $P \to Q$; we often abbreviate an object to Q, and we say that $a \in P$ *dies* in Q, or is *killed* by Q, if its image in Q is 0. For $x \in P$, the *localization* $P[-x] = P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is saturated in P^{gp} (if $mg = p - nx$ with $p \in P$, then $m(g + nx) = p + (m-1)nx \in P$, so $g + nx \in P$ by saturation of P and $g = (g + nx) - nx \in P[-x]$), and

$$L_x := P[-x]^\sharp$$

is the initial monoid under P in which x dies: a homomorphism $P \to T$ sending x to 0 extends uniquely to $P[-x]$ and then kills the units of $P[-x]$ because T is sharp. Let $B_P \subset A_P$ be the full

subcategory with objects $P \to L_x$, $x \in P$. Note $L_0 = P$.

A *presheaf* on the category A_P^{op} is a functor $(A_P^{\mathrm{op}})^{\mathrm{op}} = A_P \to \mathbf{Set}$ (or $\to \mathrm{Ab}$), hence a *covariant* functor on monoids under P; one should think of A_P^{op} as a category of affine "monoid schemes" $\operatorname{Spec} Q$. Accordingly, a family of homomorphisms $\{R \to Q_i\}$ in A_P is the same as a family of maps $\{Q_i \to R\}$ in A_P^{op}, and we refer to it as a family *over* R; fibre products in A_P^{op} over R are pushouts in A_P under R. As usual for such large sites we work in sufficiently large fixed universes, so that sheafification and injective resolutions are available.

3.2 The birhizomic topology

We recall the definitions from the question. For a monoid M, $x \in M^{\mathrm{gp}}$ and $y \in M$, write $x \prec y$ if there are integers m, n with $my \le x \le ny$. (This boundedness notation also appears in Molcho–Wise [2, Def. 2.1.3.1].) A homomorphism $f : M \to N$ is *infinitesimal* if, whenever $x \in M^{\mathrm{gp}}$ and $y \in M$ satisfy $f(x) \in N$, $x \prec y$ and $f(y) \prec f(x)$, one has $x \in M$. If $R \xrightarrow{f} Q \xrightarrow{g} N$ are homomorphisms, f is *birhizomic at* g if f is of finite presentation in **Mon** (discussed in §4) and for every factorization $gf = ph$ with $R \xrightarrow{h} M \xrightarrow{p} N$ and p infinitesimal, there is a unique $u : Q \to M$ with $pu = g$ and $uf = h$. A family $\{R \to Q_i\}_{i \in I}$ is a *birhizomic cover* if for every valuative monoid V and every homomorphism $v : R \to V$ there is a factorization of v through some $Q_i \to V$ such that $R \to Q_i$ is birhizomic at $Q_i \to V$. The question stipulates that the birhizomic covers (of objects R of A_P, all maps being under P) form a Grothendieck topology on A_P^{op}, the *birhizomic topology* J_{b}: a sieve on R is J_{b}-covering when it contains a birhizomic cover. (If one prefers to read J_{b} as the topology *generated* by the birhizomic covers, nothing below changes: Corollary 5.4 exhibits a topology τ for which every birhizomic cover is covering, and τ then contains J_{b} under either reading.) These definitions are taken from the question as stated; a related but different notion of "infinitesimal" homomorphism occurs in [2, Def. 2.1.2.9], and we use only the definition above.

Only two features of this definition enter the proof: (B1) a map birhizomic at anything is of finite presentation; and (B2) in a birhizomic cover of R, every homomorphism from R to a valuative monoid factors through a member of the family that is of finite presentation over R. The infinitesimal lifting condition is never analysed; see Remark 8.1.

3.3 The functors π^{-1} and π_*, and the main theorem

Give B_P^{op} the trivial topology, for which every presheaf is a sheaf, so $\mathrm{Sh}(B_P^{\mathrm{op}}) = \mathrm{PSh}(B_P^{\mathrm{op}})$. Let $j : B_P \hookrightarrow A_P$ be the inclusion. As in the question, $\pi^{-1} : \mathrm{Sh}(B_P^{\mathrm{op}}) \to \mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$ is the composite of the left Kan extension along j with J_{b}-sheafification, and π_* is its right adjoint. Concretely, for a presheaf F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, the left Kan extension is the presheaf

$$(\mathrm{Lan}_j F)(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a), \tag{1}$$

the colimit over the comma category B_P/Q whose objects are pairs (L_a, $L_a \to Q$ under P) [5, Tags 00VC, 00VE]; and π_* is restriction along j: for a sheaf E on A_P^{op}, $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}_{\mathrm{PSh}}(\mathrm{Lan}_j F, E) = \mathrm{Hom}(F, E \circ j)$, so $\pi_* E = E \circ j$. The same formulas define π^{-1} and π_* on abelian (pre)sheaves [5, Tag 00VF]; we check in §6 that the colimit (1) is filtered, so that the Kan extension of an abelian presheaf computed in Ab agrees with the one computed in sets. Since π_*

is left exact on abelian sheaves, it has a right derived functor $\mathrm{R}\pi_*$, and the unit of the adjunction gives a natural map $F \to \pi_*\pi^{-1}F \to \mathrm{R}\pi_*\pi^{-1}F$.

Theorem 3.1. *Let P be an arbitrary monoid (integral, saturated and sharp, as always). For every presheaf of abelian groups F on B_P^{op}, the unit*

$$F \longrightarrow \mathrm{R}\pi_*\pi^{-1}F$$

is a quasi-isomorphism of complexes of abelian presheaves on B_P^{op}, i.e. an isomorphism in the derived category of the category $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ of abelian presheaves. More precisely, the left Kan extension $\mathcal{G} = \mathrm{Lan}_j F$ of (1) is already a J_{b}-sheaf, so $\pi^{-1}F = \mathcal{G}$; the unit $F \to \pi_*\mathcal{G}$ is an isomorphism; and $H^n_{J_{\mathrm{b}}}(Q, \mathcal{G}) = 0$ for every $n > 0$ and every object Q of A_P, where $H^n_{J_{\mathrm{b}}}(Q, -)$ denotes the nth right derived functor of evaluation at Q on abelian J_{b}-sheaves (§7).

The title refers to the following reformulation: since π^{-1} is exact (the colimit (1) is filtered and sheafification is exact), it is left adjoint to $\mathrm{R}\pi_*$ between the bounded below derived categories of $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ and of $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$. Taking $E = \pi^{-1}F'[n]$ in $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}(F, \mathrm{R}\pi_*E)$ and using $F' \simeq \mathrm{R}\pi_*\pi^{-1}F'$, the theorem gives that the map $\mathrm{Ext}^n(F, F') \to \mathrm{Ext}^n(\pi^{-1}F, \pi^{-1}F')$ induced by π^{-1} (which, by naturality of the unit, is the composite of these two identifications) is bijective for all F, F' and n, the Ext groups being computed in the two categories just named.

3.4 Strategy

Cohomology for the birhizomic topology is awkward to compute directly, since its covers are defined by a lifting property. We therefore work with an auxiliary topology τ on A_P^{op} that has an explicit basis and is *finer* than J_{b}. Its generating covers are the *sign covers*

$$\{\, R \to R\langle x\rangle, \quad R \to R\langle -x\rangle \,\}, \qquad x \in R^{\mathrm{gp}},$$

where $R\langle x\rangle$ is the universal monoid under R in which x becomes ≥ 0 (§4). Geometrically, if $x = a - b$ with $a, b \in R$, these are the two charts of the blow-up of the ideal (a, b), saturated and sharpened; this chart construction is classical in toric and logarithmic geometry, see e.g. Nizioł [3, §4]. After the preliminaries on $R\langle x\rangle$ in §4, the proof has four steps.

1. *Comparison of topologies* (§5). Every birhizomic cover is a τ-cover (Corollary 5.4). By (B2) this reduces to Proposition 5.3: a family of finitely presented maps $R \to Q_i$ through which every valuative map $R \to V$ factors is refined by finitely many iterated sign covers. Since R may be infinitely generated, there is no a priori bound on the number of sign choices needed; we obtain one by a compactness argument on the space of "total precones", the saturated submonoids $D \subset R^{\mathrm{gp}}$ with $R \subset D$ and $D \cup (-D) = R^{\mathrm{gp}}$, a closed subset of $\{0,1\}^{R^{\mathrm{gp}}}$.

2. *The Kan extension is a τ-sheaf with split Mayer–Vietoris sequences* (§6). The value $\mathcal{G}(Q)$ depends only on which elements of P die in Q, so $\mathcal{G}$ is insensitive to *zero-reflecting* maps, i.e. homomorphisms f with $f^{-1}(0) = \{0\}$. Write $R_\pm = R\langle \pm x\rangle$ for the two members of a sign cover and $R_0 = R\langle x, -x\rangle$ for their "intersection" $R_+ \amalg_R R_-$. The maps from R to R_+, R_- and R_0 are zero-reflecting unless x or $-x$ already lies in R, in which case the cover contains the identity. In both cases the Mayer–Vietoris sequence $0 \to \mathcal{G}(R) \to \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \to \mathcal{G}(R_0) \to 0$ is split exact (Lemma 6.2).

3. *Acyclicity for τ* (§7). Let $H^n_\tau(R,-)$ denote the nth right derived functor of evaluation at R on abelian τ-sheaves. For a τ-sheaf $\mathcal{E}$ whose sign-cover Mayer–Vietoris sequences are exact on the right, we show $H^n_\tau(R,\mathcal{E}) = 0$ for all $n > 0$ and all R (Lemma 7.1) by induction on n: a class vanishing on both members of a sign cover vanishes (Mayer–Vietoris); every class vanishes on some τ-cover (locality of cohomology); and τ-covers are refined by trees of sign covers, along which the vanishing propagates from the leaves to the root.

4. *Descent to the coarser topology and conclusion* (§7–8). Since τ-sheafification is exact, the inclusion of τ-sheaves into J_{b}-sheaves preserves injectives. A τ-injective resolution of $\mathcal{G}$ is exact on every object by step 3, hence is also a J_{b}-injective resolution, so $H^n_{J_{\mathrm{b}}}(Q,\mathcal{G}) = 0$ for $n > 0$. Since π_* is restriction and $\mathcal{G}$ restricts to F, this is Theorem 3.1.

4 Adjoining inequalities

The basic construction is the universal way of forcing finitely many elements of R^{gp} to become nonnegative.

Definition 4.1. Let R be a monoid, $G = R^{\mathrm{gp}}$, and $S \subset G$ a subset. Put

$$R\langle S\rangle = \Big(\big(R + \textstyle\sum_{s\in S} \mathbb{N}s\big)^{\mathrm{sat}}\Big)^{\sharp}, \tag{2}$$

where the saturation is taken inside G and the sharpening is as in §3.1; thus if $C = (R+\sum_{s\in S}\mathbb{N}s)^{\mathrm{sat}}$, then $R\langle S\rangle = C/(C\cap(-C))$, a monoid with group completion $G/(C\cap(-C))$. It comes with the homomorphism $R \to R\langle S\rangle$ induced by $R \subset C$. For finite sets we write $R\langle s_1,\dots,s_k\rangle$.

Lemma 4.2 (Universal property)**.** *For every monoid T, composition with $R \to R\langle S\rangle$ identifies*

$$\mathrm{Hom}(R\langle S\rangle, T) = \{\, a : R \longrightarrow T : a^{\mathrm{gp}}(s) \in T \text{ for every } s \in S \,\}, \tag{3}$$

where $a^{\mathrm{gp}} : G \to T^{\mathrm{gp}}$ is the induced homomorphism of groups. In particular $R \to R\langle S\rangle$ is an epimorphism in **Mon***, and if S is finite it is of finite presentation.*

Proof. Let $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$. Then a^{gp} maps $R + \sum_{s\in S}\mathbb{N}s$ into T; it maps C into T because T is saturated (if $ng \in R + \sum \mathbb{N}s$ then $na^{\mathrm{gp}}(g) \in T$, so $a^{\mathrm{gp}}(g) \in T$); and it kills $C\cap(-C)$ because T is sharp. Hence a^{gp} induces $C/(C\cap(-C)) \to T$ extending a. The extension is unique, because a homomorphism out of the integral monoid $R\langle S\rangle$ is determined by the induced map on group completions, and $G \to G/(C\cap(-C))$ is surjective. Conversely, for any $R\langle S\rangle \to T$ the composite a with $R \to R\langle S\rangle$ has $a^{\mathrm{gp}}(s) \in T$, since s maps into $R\langle S\rangle$. Uniqueness of the extension says exactly that $R \to R\langle S\rangle$ is an epimorphism.

For finite presentation, write each $s \in S$ as $s = p_s - q_s$ with $p_s, q_s \in R$. Then (3) says that $R\langle S\rangle$ is obtained from R by adjoining one generator t_s and imposing the relation $q_s + t_s = p_s$, for each $s \in S$, in **Mon**: a homomorphism $R\langle S\rangle \to T$ is a homomorphism $a : R \to T$ together with elements $t_s \in T$ satisfying $a(q_s) + t_s = a(p_s)$, the $t_s = a^{\mathrm{gp}}(s)$ being unique by integrality. In other words, $R\langle S\rangle$ is the quotient, in **Mon**, of $R \oplus \mathbb{N}^S$ by the finitely many relations $q_s + t_s = p_s$. □

The question speaks of finite presentation "in the category of (integral, saturated, sharp) monoids". One may read this as a presentation by finitely many generators and relations in **Mon** (Q

is the quotient in **Mon** of $R \oplus \mathbb{N}^n$ by m relations $u_i = v_i$ with $u_i, v_i \in R \oplus \mathbb{N}^n$, so that $\mathrm{Hom}_R(Q, T)$, the set of homomorphisms under R, is the set of $t \in T^n$ satisfying the m equations obtained by evaluating u_i, v_i at $a : R \to T$ and t), or categorically. Under either reading, the property we shall use is the usual one: *for Q of finite presentation over R, the functor $\mathrm{Hom}_R(Q, -)$ on monoids under R commutes with filtered colimits.* The categorical reading is this property by definition; for the presentation reading it follows from the next lemma, because $\mathrm{Hom}_R(Q, T)$ is then the equalizer of two maps $T^n \rightrightarrows T^m$ natural in T, a finite limit of functors $T \mapsto T^k$, which commute with filtered colimits computed on underlying sets.

Lemma 4.3. *Filtered colimits exist in* **Mon** *and are computed as filtered colimits of ordinary commutative monoids, i.e. on underlying sets.*

Proof. Let $T = \varinjlim T_\alpha$ be a filtered colimit of ordinary commutative monoids, with T_α integral, saturated and sharp. Every element of T comes from some stage and every equality in T holds at some later stage; so cancellation ($u + v = u + w \Rightarrow v = w$) and sharpness ($u + v = 0 \Rightarrow u = v = 0$) pass from the T_α to T. Saturation: if $g = u - v \in T^{\mathrm{gp}}$ with $u, v \in T$ and $ng \in T$, say $nu = nv + w$ with $w \in T$, lift u, v, w and this equality to a stage T_α; there $n(u - v) = w \in T_\alpha$, so $u - v \in T_\alpha$ by saturation, and its image in T is g. Thus T lies in **Mon**, and it is then the colimit there. (Integrality and saturation are also covered by [4, Prop. I.1.3.6].) □

Two formal consequences of (3), both proved by comparing the functors $\mathrm{Hom}(-, T)$, are used repeatedly: for subsets $S, S' \subset G$ and a homomorphism $u : R \to R'$,

$$R\langle S\rangle \amalg_R R\langle S'\rangle \simeq R\langle S \cup S'\rangle, \tag{4}$$

$$R' \amalg_R R\langle S\rangle \simeq R'\langle u^{\mathrm{gp}}(S)\rangle. \tag{5}$$

Indeed, a map out of the left side of (4) to T is a map $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$ and $a^{\mathrm{gp}}(S') \subset T$, and a map out of the left side of (5) is a map $b : R' \to T$ with $(bu)^{\mathrm{gp}}(S) = b^{\mathrm{gp}}(u^{\mathrm{gp}}(S)) \subset T$. In particular $R\langle S\rangle\langle u^{\mathrm{gp}}(S')\rangle \simeq R\langle S \cup S'\rangle$ for $u : R \to R\langle S\rangle$.

The remaining two lemmas of this section are the algebraic heart of the argument: the first produces valuative monoids, the second says that adjoining a sign is harmless from the point of view of which elements become zero.

Lemma 4.4 (Extension of an order)**.** *Every monoid R embeds into a valuative monoid W with $W^{\mathrm{gp}} = R^{\mathrm{gp}}$.*

This is [2, Lemma 2.1.2.11]; we include the short proof.

Proof. Let $G = R^{\mathrm{gp}}$. Saturated sharp submonoids of G containing R form a poset under inclusion in which every chain has an upper bound, its union (saturation and sharpness are conditions on finitely many elements at a time). By Zorn's lemma choose a maximal one, C. We claim $C \cup (-C) = G$, so that $W = C$ is valuative with $W^{\mathrm{gp}} = G$.

Suppose that neither x nor $-x$ belongs to C. We show that $C' = (C + \mathbb{N}x)^{\mathrm{sat}}$ is still sharp; since C' is saturated, contains R, and contains $x \notin C$, this contradicts the maximality of C. Let $u, -u \in C'$. Then there are $m, n > 0$, $a, b \in C$ and $i, j \geq 0$ such that

$$mu = a + ix, \qquad -nu = b + jx.$$

Multiplying the first equation by n, the second by m, and adding,

$$0 = na + mb + (ni + mj)x.$$

If $ni+mj > 0$, then $-(ni+mj)x = na+mb \in C$, and saturation of C gives $-x \in C$, a contradiction. Otherwise $i = j = 0$, so mu and $-nu$ lie in C; then mnu and $-mnu$ lie in C, sharpness of C gives $mnu = 0$, and torsion-freeness of G gives $u = 0$. □

Call a homomorphism of monoids $f : Q \to Q'$ *zero-reflecting* if $f^{-1}(0) = \{0\}$. (Between sharp monoids this is the condition that f reflects units, called *local* in [4, p. 15].)

Lemma 4.5. *Let $x \in R^{\mathrm{gp}}$, and set*

$$R_+ = R\langle x\rangle, \qquad R_- = R\langle -x\rangle, \qquad R_0 = R\langle x, -x\rangle.$$

1. *If neither x nor $-x$ lies in R, the three maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting.*
2. *If $x \in R$, then $R_+ = R$, and $R_0 = R_-$ with $R_- \to R_0$ the identity and $R_+ \to R_0$ the map $R \to R_-$. Symmetrically, if $-x \in R$, then $R_- = R$ and $R_0 = R_+$.*

Proof. (2) If $x \in R$, then $R + \mathbb{N}x = R$, which is saturated and sharp, so $R_+ = R$; and $R\langle x, -x\rangle = R\langle -x\rangle$ by (2), since $\mathbb{N}x \subset R$. The description of the maps follows from the uniqueness in Lemma 4.2. The case $-x \in R$ is obtained by replacing x with $-x$.

(1) Since $R \to R_0$ factors through $R \to R_+$ and through $R \to R_-$ (Lemma 4.2), it suffices to treat R_0. Before sharpening, R_0 is $C = (R + \mathbb{Z}x)^{\mathrm{sat}}$. If $r \in R$ maps to zero in $C^\sharp$, then $-r \in C$, so

$$-nr = a + kx \qquad (n > 0,\ a \in R,\ k \in \mathbb{Z}).$$

If $k > 0$, then $-kx = a + nr \in R$, and saturation gives $-x \in R$; if $k < 0$, then $(-k)x = a + nr \in R$ gives $x \in R$. Both are excluded. If $k = 0$, then nr and $-nr$ lie in R, so $nr = 0$ by sharpness and $r = 0$ by torsion-freeness. □

5 A topology finer than the birhizomic topology

Definition 5.1. For an object R of A_P and $x \in R^{\mathrm{gp}}$, the *sign cover* of R at x is the family

$$\{\, R \longrightarrow R\langle x\rangle, \quad R \longrightarrow R\langle -x\rangle \,\} \qquad \text{(written in the monoid direction).} \tag{6}$$

Let τ be the Grothendieck topology on A_P^{op} generated by the sign covers, i.e. the smallest topology for which the sieve generated by every sign cover is covering.

By (5), the base change of the sign cover at x along $u : R \to R'$ (a fibre product in A_P^{op}) is the sign cover of R' at $u^{\mathrm{gp}}(x)$: *sign covers are stable under base change*. This gives τ an explicit description. A *tree family* over R is the family of leaves $\{R \to T_\ell\}_{\ell \in \Lambda}$ of a finite binary tree with root R in which the two children of each internal vertex M form a sign cover of M ($R \to T_\ell$ being the composite along the path from the root); its *depth* is the depth of the tree, the trivial tree (depth 0) giving $\{\mathrm{id}_R\}$. Base change along $u : R \to R'$ takes a tree family over R to a tree family over R' of the same depth (apply (5) vertex by vertex), and replacing each leaf of a tree

family by a tree family over that leaf (grafting) gives again a tree family. Consequently the sieves containing a tree family form a Grothendieck topology (the three axioms—maximal sieve, stability under base change, local character, i.e. transitivity—follow from the trivial tree, base change and grafting respectively), in which sign covers are covering; conversely, in any topology in which sign covers are covering, every tree family generates a covering sieve, by induction on the depth using the local character axiom. Hence

$$\text{a sieve on } R \text{ is } \tau\text{-covering if and only if it contains a tree family over } R. \tag{7}$$

For τ, or any other topology J on A_P^{op}, we say that a family over R is a *J-cover* if the sieve it generates is J-covering. Thus a family over R is a τ-cover as soon as it is *refined* by a tree family, meaning that every leaf T_ℓ receives a map under R from some member of the family.

Members of tree families are epimorphisms, being composites of the epimorphisms $M \to M\langle \pm x\rangle$ (Lemma 4.2). Recall that a presheaf $\mathcal{F}$ on A_P^{op} satisfies the *sheaf condition* for a family $\{R \to T_\ell\}_{\ell\in\Lambda}$ if $\mathcal{F}(R) \to \prod_\ell \mathcal{F}(T_\ell)$ is injective with image the *compatible* families (s_ℓ), those for which s_ℓ and s_m have the same image in $\mathcal{F}(T_\ell \amalg_R T_m)$ for all ℓ, m (fibre products in A_P^{op} being pushouts in A_P); when $R \to T_\ell$ is an epimorphism, $T_\ell \amalg_R T_\ell = T_\ell$ and the condition for $\ell = m$ is vacuous. For a sign cover, compatibility of (s_+, s_-) means that s_+ and s_- have the same image in $\mathcal{F}(R\langle x\rangle \amalg_R R\langle -x\rangle) = \mathcal{F}(R\langle x, -x\rangle)$ (4).

Lemma 5.2 (Sheaf criterion)**.** *A presheaf $\mathcal{F}$ of sets or of abelian groups on A_P^{op} is a τ-sheaf as soon as it satisfies the sheaf condition for every sign cover* (6)*.*

Proof. Step A: $\mathcal{F}$ satisfies the sheaf condition for every tree family, over every object. We induct on the depth d; depth 0 is trivial and depth 1 is the hypothesis. Let $\{R \to T_\ell\}$ have depth $d+1 \geq 2$, let $R_\pm$ be the children of the root and $\Lambda = \Lambda_+ \sqcup \Lambda_-$ the leaves of the two subtrees, which are tree families over $R_\pm$ of depth $\leq d$. Let (s_ℓ) be compatible. For $\ell, m \in \Lambda_+$, restricting along the canonical map $T_\ell \amalg_R T_m \to T_\ell \amalg_{R_+} T_m$ shows that $(s_\ell)_{\ell\in\Lambda_+}$ is compatible over R_+, so by induction there is a unique $s_+ \in \mathcal{F}(R_+)$ restricting to s_ℓ for all $\ell \in \Lambda_+$; similarly s_-. We claim that s_+ and s_- have the same image in $\mathcal{F}(R_0)$, $R_0 = R_+ \amalg_R R_-$. The base change of the R_+-subtree along $R_+ \to R_0$ is a tree family over R_0 of depth $\leq d$ with leaves $T_\ell \amalg_{R_+} R_0 = T_\ell \amalg_R R_-$ ($\ell \in \Lambda_+$), so by the injectivity part of the induction hypothesis it suffices to compare the restrictions of s_+ and s_- to each $T_\ell \amalg_R R_-$; these are the restrictions of s_ℓ and of s_-. In turn, the base change of the R_--subtree along $R_- \to T_\ell \amalg_R R_-$ is a tree family of depth $\leq d$ with leaves $T_\ell \amalg_R T_m$ ($m \in \Lambda_-$), on which s_ℓ and s_- restrict to the images of s_ℓ and s_m, which agree by compatibility; injectivity once more proves the claim. By the sheaf condition for the sign cover $\{R \to R_\pm\}$ there is a unique $s \in \mathcal{F}(R)$ restricting to $s_\pm$, hence to all s_ℓ. If $s' \in \mathcal{F}(R)$ also restricts to all s_ℓ, then $s'|_{R_\pm} = s_\pm$ by the uniqueness in the induction hypothesis, so $s' = s$.

Step B: $\mathcal{F}$ is a τ-sheaf (this is the usual comparison between sheaves for a site and for its associated topology, cf. [5, Tag 00ZC]; we include the argument since tree families, not being closed under isomorphism, do not form a site in the sense of that tag). Let Σ be a τ-covering sieve on R; by (7) it contains a tree family $\{R \to T_\ell\}$. A compatible family on Σ is a collection of sections $s_V \in \mathcal{F}(V)$, one for each arrow $(R \to V) \in \Sigma$ (sections are indexed by arrows; we write s_V when the arrow is clear), with $s_{V'} = s_V|_{V'}$ for every $V \to V'$ under R, and we must show that $s \mapsto (s|_V)_V$ is a bijection from $\mathcal{F}(R)$ onto such collections. Injectivity follows from Step A for $\{R \to T_\ell\}$. Given (s_V), the sections s_{T_ℓ} are compatible (both s_{T_ℓ} and s_{T_m} restrict to $s_{T_\ell \amalg_R T_m}$), so Step A gives $s \in \mathcal{F}(R)$ with $s|_{T_\ell} = s_{T_\ell}$ for all ℓ. For $(R \to V) \in \Sigma$, the sections $s|_V$ and s_V both restrict to $s_{V \amalg_R T_\ell}$ on each leaf of the base-changed tree family $\{V \to V \amalg_R T_\ell\}$, hence are equal by Step A. $\square$

Our goal in this section is to show that τ is finer than J_{b}. The next proposition concerns finitely presented families that "see" all valuative monoids; by (B2), it applies to birhizomic covers. It contains the compactness argument needed for possibly infinitely generated R.

Proposition 5.3. *Let R be an object of A_P and let $\{R \to Q_i\}_{i\in I}$ be a family of maps of finite presentation such that every homomorphism $R \to V$ to a valuative monoid V factors through some $R \to Q_i$. Then the family is refined by a tree family; in particular it is a τ-cover.*

Proof. Step 1: a compact space of orders. Let $G = R^{\mathrm{gp}}$, and let $\mathcal{V}(R)$ be the set of saturated submonoids $D \subset G$ with

$$R \subset D, \qquad D \cup (-D) = G.$$

Such a D is a "total precone": it induces a total preorder on G, but it need not be sharp, and it is essential to allow this: the precones D_ϵ produced in Step 5 below contain the kernel of a surjection $G \to T_\epsilon^{\mathrm{gp}}$ and are not sharp in general. Identify D with its characteristic function and regard $\mathcal{V}(R)$ as a subset of the product $\{0,1\}^G$, compact by Tychonoff's theorem. It is closed: each of the conditions $r \in D$ (for $r \in R$), $0 \in D$, "$g, h \in D \Rightarrow g + h \in D$", "$ng \in D \Rightarrow g \in D$" and "$g \in D$ or $-g \in D$" involves finitely many coordinates, and $\mathcal{V}(R)$ is the intersection of these closed conditions. Hence $\mathcal{V}(R)$ is compact. (The same device, embedding a set of valuation-like objects into a Boolean product, appears in [1, Lemma 5.3.6].) For finite $S \subset G$, the subset

$$U_S = \{D \in \mathcal{V}(R) : S \subset D\}$$

is open and closed, being defined by finitely many coordinates.

Step 2: the valuative monoid of a precone as a filtered colimit. For $D \in \mathcal{V}(R)$, write $H_D = D \cap (-D)$ and $V_D = D/H_D = D^\sharp$. Then V_D is a monoid (integral, saturated, sharp, cf. §3.1), valuative because $D \cup (-D) = G$, and a monoid under R via $R \subset D \to V_D$. We claim that, in the category of monoids under R,

$$V_D = \varinjlim_{\substack{S \subset D \\ S \text{ finite}}} R\langle S\rangle, \tag{8}$$

the colimit being over the directed set of finite subsets of D, with transition maps $R\langle S\rangle \to R\langle S'\rangle$ for $S \subset S'$ given by Lemma 4.2. Indeed, by (3) a compatible family of maps $R\langle S\rangle \to T$ under $a : R \to T$ is the same as the condition $a^{\mathrm{gp}}(D) \subset T$. Given this condition, $a^{\mathrm{gp}}|_D : D \to T$ kills H_D (as T is sharp) and so factors uniquely through V_D; conversely a map $V_D \to T$ under R composed with $D \to V_D$ shows $a^{\mathrm{gp}}(D) \subset T$. So V_D represents the functor $\varprojlim_S \mathrm{Hom}_R(R\langle S\rangle, -)$, which is the claim.

Step 3: finite presentation yields finite stages. Fix $D \in \mathcal{V}(R)$. By hypothesis the valuative map $R \to V_D$ factors as $R \to Q_i \to V_D$ for some i. Since (8) is a filtered colimit of monoids under R and $\mathrm{Hom}_R(Q_i, -)$ commutes with such colimits (finite presentation; see the discussion following Lemma 4.2), the map $Q_i \to V_D$ factors as

$$Q_i \longrightarrow R\langle S_D\rangle \longrightarrow V_D$$

under R, for some finite $S_D \subset D$. The clopen sets U_{S_D}, $D \in \mathcal{V}(R)$, cover $\mathcal{V}(R)$, since $D \in U_{S_D}$. By compactness choose $D_1, \ldots, D_m$ with $\mathcal{V}(R) = U_{S_1} \cup \cdots \cup U_{S_m}$, where $S_j = S_{D_j}$, and retain the maps

$$Q_{i_j} \longrightarrow R\langle S_j\rangle \qquad (j = 1, \ldots, m) \tag{9}$$

under R. Put $S = S_1 \cup \cdots \cup S_m$, a finite subset of G.

Step 4: the sign tree. Enumerate $S = \{s_1, \dots, s_k\}$ and form the binary tree of depth k whose vertex at level ℓ with sign history $\epsilon(s_1), \dots, \epsilon(s_\ell)$ is $R\langle \epsilon(s_1)s_1, \dots, \epsilon(s_\ell)s_\ell \rangle$; by (5) and (4), the sign cover of this vertex at the image of $s_{\ell+1}$ has as members the two vertices at the next level. The leaves are

$$T_\epsilon = R\langle\, \epsilon(s)s : s \in S \,\rangle, \qquad \epsilon : S \longrightarrow \{1, -1\}, \tag{10}$$

so $\{R \to T_\epsilon\}_\epsilon$ is a tree family. It remains to show that each $R \to T_\epsilon$ factors through some $R\langle S_j \rangle$; composing with (9) then shows that the tree family refines $\{R \to Q_i\}$.

Step 5: each leaf factors through some $R\langle S_j \rangle$. The idea is to attach to each leaf T_ϵ a precone $D_\epsilon \in \mathcal{V}(R)$ extending the inequalities $\epsilon(s)s \geq 0$ that define T_ϵ; the finite subcover gives $S_j \subset D_\epsilon$ for some j, and we then check that the elements of S_j are nonnegative already in T_ϵ. Fix ϵ. By Lemma 4.4, embed T_ϵ into a valuative monoid W_ϵ with $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$. Let $D_\epsilon \subset G$ be the inverse image of W_ϵ under the group homomorphism $G \to T_\epsilon^{\mathrm{gp}} = W_\epsilon^{\mathrm{gp}}$. Then $D_\epsilon \in \mathcal{V}(R)$: it contains R (which maps into $T_\epsilon \subset W_\epsilon$), it is saturated (as the inverse image of a saturated monoid under a group homomorphism), and $D_\epsilon \cup (-D_\epsilon) = G$ because $W_\epsilon \cup (-W_\epsilon) = W_\epsilon^{\mathrm{gp}}$. Choose j with $D_\epsilon \in U_{S_j}$, i.e. $S_j \subset D_\epsilon$.

We check that every $s \in S_j$ becomes nonnegative in T_ϵ. Let $\bar{s} \in T_\epsilon^{\mathrm{gp}}$ be the image of s. If $\epsilon(s) = 1$, then $\bar{s} \in T_\epsilon$ by construction of T_ϵ. If $\epsilon(s) = -1$, then $-\bar{s} \in T_\epsilon \subset W_\epsilon$, while $s \in D_\epsilon$ says $\bar{s} \in W_\epsilon$; sharpness of W_ϵ gives $\bar{s} = 0$ in $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$, so again $\bar{s} \in T_\epsilon$. (Here it matters that W_ϵ has the *same* group completion as T_ϵ: a map to an arbitrary valuative monoid would only give $\bar{s} \mapsto 0$ there.) By Lemma 4.2 there is a map $R\langle S_j \rangle \to T_\epsilon$ under R, as required. □

Corollary 5.4. *Every birhizomic covering sieve on A_P^{op} is τ-covering; that is, J_{b} is coarser than τ, $J_{\mathrm{b}} \subset \tau$. Consequently every τ-sheaf is a J_{b}-sheaf.*

Proof. Let $\{R \to Q_i\}_{i \in I}$ be a birhizomic cover and let $I' \subset I$ be the set of indices for which $R \to Q_i$ is of finite presentation. By (B1) and the definition of a birhizomic cover, every homomorphism from R to a valuative monoid factors through some Q_i with $i \in I'$. Proposition 5.3 applied to $\{R \to Q_i\}_{i \in I'}$ shows that the sieve generated by this subfamily, hence also the sieve generated by the whole family, is τ-covering. Every J_{b}-covering sieve contains a birhizomic cover and is therefore τ-covering. The last assertion follows, since a τ-sheaf satisfies the sheaf condition for every τ-covering sieve, in particular for every J_{b}-covering one. □

6 The Kan extension and its Mayer–Vietoris sequences

Fix a presheaf of abelian groups F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, and let $\mathcal{G} = \mathrm{Lan}_j F$ be its left Kan extension (1) to A_P, a presheaf of abelian groups on A_P^{op}:

$$\mathcal{G}(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a).$$

We first make the indexing category explicit. By the universal property of $L_a = P[-a]^\sharp$, a map $L_a \to Q$ under P exists if and only if the image of a in Q is zero, and it is then unique. In particular B_P is a *thin* category (a preorder): there is at most one map $L_a \to L_b$, and it exists if and only if $a \mapsto 0$ in L_b. So B_P/Q is the preorder of elements $a \in P$ that die in Q (elements giving the same object of B_P being identified), with $a \leq b$ when a dies in L_b. It is nonempty, since it contains

$L_0 = P$, and directed: if a and b die in Q, so does $a+b$, and $a, b \mapsto 0$ in L_{a+b} because the images of a and b in the sharp monoid L_{a+b} sum to zero. Thus (1) is a filtered colimit, and since filtered colimits of abelian groups are computed on underlying sets, the Kan extension of F in abelian presheaves is the same as in presheaves of sets, as announced in the introduction.

Two consequences will be used.

Lemma 6.1. *1. For $Q = L_a$ in B_P, the comma category B_P/L_a has the terminal object (L_a, id), so the unit of the Kan extension is an isomorphism*

$$F(L_a) \xrightarrow{\sim} \mathcal{G}(L_a). \tag{11}$$

2. If $Q \to Q'$ is a zero-reflecting map under P, then an element $a \in P$ dies in Q if and only if it dies in Q', so the comma categories B_P/Q and B_P/Q' coincide, and the restriction map

$$\mathcal{G}(Q) \xrightarrow{\sim} \mathcal{G}(Q') \tag{12}$$

is an isomorphism (it is the identity of the common colimit).

Proof. (1) Every object $(L_b, L_b \to L_a)$ of B_P/L_a has the given map as its unique map to (L_a, id), by thinness; a colimit over a category with a terminal object is the value at that object, and the unit is the corresponding structure map. (2) An element killed in Q is killed in Q'; conversely, if $a \mapsto 0$ in Q', then the image of a in Q is killed by the zero-reflecting map $Q \to Q'$, so it is 0. The restriction map $\mathcal{G}(Q) \to \mathcal{G}(Q')$ is induced by the identity of the diagrams, hence is the identity. □

In words, $\mathcal{G}(Q)$ depends only on the set $\{a \in P : a \text{ dies in } Q\}$ of elements of P killed by Q (a *face* of P: a submonoid containing a and b whenever it contains $a+b$), and $\mathcal{G}$ is "locally constant" along zero-reflecting maps. Combined with Lemma 4.5, which says that sign covers consist of zero-reflecting maps except in the degenerate case, this gives the sheaf property and more.

Lemma 6.2. *For every sign cover (6), with $R_\pm$ and R_0 as in Lemma 4.5, the sequence*

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \longrightarrow \mathcal{G}(R_0) \longrightarrow 0, \tag{13}$$

whose first map is the pair of restrictions and whose second map is the difference of the two restrictions to R_0, is exact, and in fact split. Moreover $\mathcal{G}$ is a τ-sheaf, hence (Corollary 5.4) a J_{b}-sheaf.

Proof. Suppose first that neither x nor $-x$ lies in R. By Lemma 4.5(1) the maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting, so by (12) the three restriction maps out of $\mathcal{G}(R)$ are isomorphisms; since $\mathcal{G}(R) \to \mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ composes to the isomorphism $\mathcal{G}(R) \to \mathcal{G}(R_0)$, the maps $\mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ are isomorphisms too. Identifying all four groups with $A = \mathcal{G}(R)$ through these isomorphisms, every restriction map becomes the identity of A and the sequence becomes

$$0 \longrightarrow A \xrightarrow{a \mapsto (a,a)} A \oplus A \xrightarrow{(a,b) \mapsto a-b} A \longrightarrow 0,$$

which is exact, and split by $a \mapsto (a, 0)$.

If $x \in R$, then by Lemma 4.5(2) $R_+ = R$, $R_0 = R_-$, the map $R_- \to R_0$ is the identity and $R_+ \to R_0$ is $R \to R_-$. Writing $A = \mathcal{G}(R)$, $B = \mathcal{G}(R_-)$ and $\rho : A \to B$ for the restriction, the sequence reads

$$0 \longrightarrow A \xrightarrow{a \mapsto (a, \rho(a))} A \oplus B \xrightarrow{(a,b) \mapsto \rho(a) - b} B \longrightarrow 0,$$

which is exact, split by $(a,b)\mapsto a$, for any homomorphism ρ. The case $-x\in R$ is the same with the roles of R_+ and R_- exchanged.

For the sheaf property, by Lemma 5.2 it suffices to check the sheaf condition for sign covers, and as noted before that lemma this condition says that $\mathcal{G}(R)$ is the equalizer of the two restriction maps $\mathcal{G}(R_+)\times\mathcal{G}(R_-)\rightrightarrows\mathcal{G}(R_0)$, which is exactness of (13) at the first two terms. □

7 Acyclicity, and passage to the coarser topology

For a Grothendieck topology J on A_P^{op}, an object R and an abelian J-sheaf $\mathcal{E}$, we write $H^n_J(R,\mathcal{E})$ for the nth right derived functor of $\mathcal{E}\mapsto\mathcal{E}(R)$ on the abelian category $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}},J)$ of abelian J-sheaves, computed by injective resolutions. We shall use two standard facts. First, the Mayer–Vietoris sequence [5, Tag 0EVY] for a cover $\{R_\pm\to R\}$ in A_P^{op} by two monomorphisms ($R\to R_\pm$ being epimorphisms in A_P) with fibre product R_0; we spell out the version we need in the proof below. Second, *locality of cohomology* [5, Tag 03P1]: for $n>0$, every class $\xi\in H^n_J(R,\mathcal{E})$ vanishes on the members of some J-cover of R (in the sense of §5). (Explicitly: if $\mathcal{E}\to\mathcal{I}^\bullet$ is an injective resolution and $z\in\mathcal{I}^n(R)$ a cocycle representing ξ, the exactness of $\mathcal{I}^\bullet$ as a complex of *sheaves* in degree $n>0$ says that z is locally a coboundary.)

Lemma 7.1. *Let $\mathcal{E}$ be an abelian τ-sheaf on A_P^{op} such that for every sign cover* (6)*, with $R_\pm$ and R_0 as in Lemma 4.5, the difference map*

$$\mathcal{E}(R_+)\oplus\mathcal{E}(R_-)\longrightarrow\mathcal{E}(R_0) \tag{14}$$

is surjective. Then $H^n_\tau(R,\mathcal{E})=0$ for all $n>0$ and all objects R of A_P.

Proof. Mayer–Vietoris. For an object Y of A_P let h_Y be the presheaf represented by Y on A_P^{op}, so $h_Y(X)=\mathrm{Hom}_{A_P}(Y,X)$, and let $\mathbb{Z}_\tau[Y]$ be the τ-sheaf associated with the free abelian presheaf $\mathbb{Z}[h_Y]$; then $\mathrm{Hom}(\mathbb{Z}_\tau[Y],\mathcal{E})=\mathcal{E}(Y)$ for every abelian τ-sheaf $\mathcal{E}$ [5, Tag 03AB], so that $H^n_\tau(Y,\mathcal{E})=\mathrm{Ext}^n(\mathbb{Z}_\tau[Y],\mathcal{E})$, the Ext group being computed in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}},\tau)$ (apply $\mathrm{Hom}(\mathbb{Z}_\tau[Y],-)$ to an injective resolution of $\mathcal{E}$). Fix a sign cover of R at x. For every X, the universal property (3) identifies $h_{R_\pm}(X)$ with subsets of $h_R(X)$, with intersection $h_{R_0}(X)$ by (4). The elementary exact sequence $0\to\mathbb{Z}[U\cap U']\to\mathbb{Z}[U]\oplus\mathbb{Z}[U']\to\mathbb{Z}[U\cup U']\to0$ of free abelian groups on two subsets of a set therefore gives an exact sequence of presheaves

$$0\longrightarrow\mathbb{Z}[h_{R_0}]\longrightarrow\mathbb{Z}[h_{R_+}]\oplus\mathbb{Z}[h_{R_-}]\longrightarrow\mathbb{Z}[h_R]$$

whose cokernel is $\mathbb{Z}[h_R]/\mathbb{Z}[h_{R_+}\cup h_{R_-}]$. This cokernel has zero τ-sheafification: a section of h_R over X is a map $f:R\to X$, and on the two members of the sign cover of X at $f^{\mathrm{gp}}(x)$, which are $X\amalg_R R_\pm$ by (5), the restrictions of f lie in h_{R_+} and h_{R_-} respectively. Since sheafification is exact we obtain a short exact sequence of τ-sheaves

$$0\longrightarrow\mathbb{Z}_\tau[R_0]\longrightarrow\mathbb{Z}_\tau[R_+]\oplus\mathbb{Z}_\tau[R_-]\longrightarrow\mathbb{Z}_\tau[R]\longrightarrow0,$$

and applying $\mathrm{Ext}^\bullet(-,\mathcal{E})$ the long exact sequence

$$\begin{aligned}0\to\mathcal{E}(R)\to\mathcal{E}(R_+)\oplus\mathcal{E}(R_-)\to\mathcal{E}(R_0)\to H^1_\tau(R,\mathcal{E})\to\cdots\\ \cdots\to H^{n-1}_\tau(R_0,\mathcal{E})\to H^n_\tau(R,\mathcal{E})\to H^n_\tau(R_+,\mathcal{E})\oplus H^n_\tau(R_-,\mathcal{E})\to H^n_\tau(R_0,\mathcal{E})\to\cdots\end{aligned} \tag{15}$$

This is the Mayer–Vietoris sequence of [5, Tag 0EVY] for the present situation.

Injectivity on the members of a sign cover. We prove the lemma by induction on $n \geq 1$, simultaneously for all objects R. Consider the statement $(*)_n$ that, for every sign cover of every object R, the map

$$H^n_\tau(R, \mathcal{E}) \longrightarrow H^n_\tau(R_+, \mathcal{E}) \oplus H^n_\tau(R_-, \mathcal{E})$$

is injective. For $n = 1$, $(*)_1$ follows from (15) and the assumed surjectivity of (14), which makes the connecting map $\mathcal{E}(R_0) \to H^1_\tau(R, \mathcal{E})$ zero. For $n > 1$, if $H^{n-1}_\tau(-, \mathcal{E}) = 0$ on all objects, then in particular $H^{n-1}_\tau(R_0, \mathcal{E}) = 0$ and (15) gives $(*)_n$.

> **Reviewer 1**
>
> This is probably the most important moment of the proof. It appears correct, but is dismissed a little too informally.

From the leaves to the root. Having established $(*)_n$ (unconditionally for $n = 1$, and under the induction hypothesis for $n > 1$), let $\xi \in H^n_\tau(R, \mathcal{E})$. By locality of cohomology, ξ vanishes on the members of some τ-cover of R. By (7), the sieve generated by this family contains a tree family $\{R \to T_\ell\}$ (each T_ℓ receives a map under R from a member on which ξ vanishes), so $\xi|_{T_\ell} = 0$ for every leaf. Now argue upward through the tree: if M is an internal vertex with children M_+, M_- forming a sign cover of M, and $\xi|_{M_+} = 0 = \xi|_{M_-}$, then $\xi|_M = 0$ by $(*)_n$ applied to the sign cover of M. Induction on the height of a vertex gives $\xi|_R = \xi = 0$. Hence $H^n_\tau(R, \mathcal{E}) = 0$ for all R, which completes the induction on n. □

Corollary 7.2. *Let $\mathcal{G} = \mathrm{Lan}_j F$ be the Kan extension of §6. For every object R of A_P and every $n > 0$, $H^n_\tau(R, \mathcal{G}) = 0$.*

Proof. By Lemma 6.2, $\mathcal{G}$ is a τ-sheaf whose maps (14) are surjective. Apply Lemma 7.1. □

We now transfer this vanishing to the coarser topology J_{b}. Some care is needed, since the inclusion of τ-sheaves into J_{b}-sheaves is not exact in general (cokernels involve different sheafifications); what does transfer is injectivity of objects, and Corollary 7.2 makes a τ-injective resolution exact already as a complex of presheaves.

Proposition 7.3. *For $\mathcal{G} = \mathrm{Lan}_j F$, every object R of A_P and every $n > 0$, $H^n_{J_{\mathrm{b}}}(R, \mathcal{G}) = 0$.*

Proof. Since $J_{\mathrm{b}} \subset \tau$ (Corollary 5.4), the identity of A_P^{op} is a morphism of sites $(A_P^{\mathrm{op}}, \tau) \to (A_P^{\mathrm{op}}, J_{\mathrm{b}})$ whose direct image is the inclusion

$$\iota : \mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, \tau) \longrightarrow \mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$$

and whose inverse image is τ-sheafification, an exact functor [5, Tag 0EWK]. A right adjoint whose left adjoint is exact preserves injectives [5, Tag 015Z]; hence ι carries injective τ-sheaves to injective J_{b}-sheaves.

Choose an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, \tau)$. By Corollary 7.2, for every object R the augmented complex of abelian groups

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{I}^0(R) \longrightarrow \mathcal{I}^1(R) \longrightarrow \cdots \tag{16}$$

is exact. So $\mathcal{G} \to \mathcal{I}^\bullet$ is exact as a complex of presheaves, hence exact as a complex of J_{b}-sheaves (kernels of maps of sheaves are computed in presheaves, and a presheaf image that is already a sheaf

equals the sheaf image). Its terms $\iota\mathcal{I}^k$ are injective J_{b}-sheaves by the preceding paragraph. Thus $\mathcal{G} \to \iota\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{G}$ in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$, and $H^n_{J_{\mathrm{b}}}(R, \mathcal{G})$ is the nth cohomology of the same complex (16), which vanishes for $n > 0$. □

8 Proof of Theorem 3.1

Recall from §3.3 that $\pi^{-1}F$ is the J_{b}-sheafification of the left Kan extension $\mathcal{G} = \mathrm{Lan}_j F$, and that π_* is restriction along $j : B_P \hookrightarrow A_P$.

$\pi^{-1}F = \mathcal{G}$. By Lemma 6.2, $\mathcal{G}$ is already a J_{b}-sheaf, so no sheafification is needed and $\pi^{-1}F = \mathcal{G}$.

The unit in degree zero. The unit $F \to \pi_*\pi^{-1}F = \mathcal{G} \circ j$ evaluated at L_a is the Kan extension unit $F(L_a) \to \mathcal{G}(L_a)$, an isomorphism by (11).

Vanishing of $R^n\pi_*$ for $n > 0$. Let $\mathcal{G} \to \mathcal{I}^\bullet$ be an injective resolution in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$. By definition $\mathrm{R}\pi_*\mathcal{G}$ is represented by the complex of presheaves $\pi_*\mathcal{I}^\bullet = \mathcal{I}^\bullet \circ j$ on B_P^{op}. Since B_P^{op} carries the trivial topology, evaluation at any object L_a is an exact functor on $\mathrm{Ab}\,\mathrm{Sh}(B_P^{\mathrm{op}}) = \mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$, so

$$(R^n\pi_*\mathcal{G})(L_a) = H^n\big(\mathcal{I}^\bullet(L_a)\big) = H^n_{J_{\mathrm{b}}}(L_a, \mathcal{G}) = 0 \qquad (n > 0)$$

by Proposition 7.3. As the L_a are all the objects of B_P, the cohomology presheaves $R^n\pi_*\mathcal{G}$ vanish for $n > 0$, and $R^0\pi_*\mathcal{G} = \pi_*\mathcal{G} \cong F$ by the previous paragraph. Hence $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism. This proves Theorem 3.1, including the three "more precise" assertions. □

Remark 8.1 (Where the hypotheses are used). The infinitesimal lifting condition in the definition of a birhizomic cover is never analysed: Proposition 5.3 shows that *any* family of finitely presented maps through which every valuative map factors is a τ-cover, and this is all that is used. Finite presentation enters exactly once, in Step 3 of that proof, to factor $Q_i \to V_D$ through a finite stage of the colimit (8); the colimit identity itself needs no finiteness. Finite generation of P is never used, which is why the compactness argument on $\mathcal{V}(R)$ is needed in place of a dimension count. Finally, the argument of Proposition 7.3 shows that $\mathcal{G}$ is acyclic for every topology coarser than τ, the birhizomic topology being one instance.

Remark 8.2 (Two small examples). For $R = \mathbb{N}^2$ and $x = e_1 - e_2$, neither sign lies in R; the members of the sign cover are $\mathbb{N}^2\langle\pm(e_1 - e_2)\rangle$, the two charts of the blow-up of the origin, and $R_0 = \mathbb{N}^2\langle x, -x\rangle \simeq \mathbb{N}$ via $(a, b) \mapsto a + b$. None of these maps kills a nonzero element of $\mathbb{N}^2$, so $\mathcal{G}$ takes the same value on all four monoids, as in the first case of Lemma 6.2. For $R = \mathbb{N}$ and $x = 1$ the sign cover is $\{\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to 0\}$ with $R_0 = 0$, and (13) is $0 \to \mathcal{G}(\mathbb{N}) \to \mathcal{G}(\mathbb{N}) \oplus \mathcal{G}(0) \to \mathcal{G}(0) \to 0$. Deeper in a sign tree a leaf may kill nonzero elements of P; this is harmless, as Lemma 4.5 is applied at each internal vertex.

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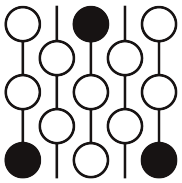
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[5] The Stacks Project Authors, *The Stacks Project*, 2026. Tags 00VC, 00VE, 00VF (Kan extension along a functor and its adjoint), 00ZC (the topology defined by a site), 03AB (free abelian sheaves), 015Z (exact left adjoints and injectives), 03P1 (locality of cohomology), 0EVY (Mayer–Vietoris on a site), 0EWK (comparing two topologies).

Review of Problem 1 – ProofCouncil

First Proof Third Batch
Reviewer 2

October 8–9, 2026



1 Review

The submission was correct, clear, and thorough. The tree formalism was somewhat more than required for the proof, but I appreciated that its use resulted in somewhat better clarity for the structure of the refining covers. I found its solution to be the clearest of all the solutions I read.

Certain passages were somewhat misstated, vague, or problematic. Its "zero-reflecting" terminology was not standard in the literature – usually, this property is dubbed either "local" or "sharp." This reflects a greater familiarity with, e.g., category-theoretic naming conventions. The claim of finite presentation in Lemma 4.2 depends on which category it is asserted. I believed this was an error, but the paragraph *afterwards* clarifies that finite presentation is meant in the correct category. More natural would have been to describe the assertion within Lemma 4.2. I also found two of the neighboring sentences needlessly opaque.

Similarly, Lemma 7.1 contains an unfortunate use of wording which I initially believed to be a mathematical error. In "injectivity on the members of a sign cover," the inductive step is for the entire lemma, rather than the present claim $(*)_n$. Reading it as the latter, it seems to make an unnecessarily circular argument. This was the most frustrating part of the draft as a reader.

The writing would assert that vertices of a graph form a "sign cover," which has only one obvious interpretation but is somewhat careless. Same for "sign cover of R at x" in Lemma 7.1. The statement in equation (7) should have been a lemma. It cited (B1) in Corollary 5.4, which was not a good choice as it referred both to a general principal instead of a result and would have been better phrased in natural language. In defining the preorder at the start of §6, it refrained from using standard terminology $\preceq$ for the preorder or citing standard facts for that relation. I would assign this submission a .8 in exposition if not for the requirement that exposition scores lie in $\mathbb{Z}$.

I was very impressed by Step B in Lemma 5.2 in that it reveals a great deal of thoroughness in the solution. However, this technical point could have been easily avoided in allowing isomorphisms in the definition of the covers.

It was thorough in citing both the results that were used, but also related constructions and ideas in the literature that were not used. I found it to be the most generous in this way among the solutions.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected

- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 1

> **Reviewer 2**
>
> Did a good job of citing both for results used, parallel arguments, or even similar constructions/ideas in the literature.

2 Submission with inline comments

Abstract

Let P be an integral, saturated, sharp commutative monoid, let A_P be the category of monoids under P, equipped with the birhizomic topology on A_P^{op}, and let $B_P \subset A_P$ be the full subcategory of the sharpened localizations $P \to P[-x]^\sharp$, $x \in P$, with the trivial topology (all definitions are recalled in §1). The inclusion induces an inverse image functor π^{-1} from presheaves on B_P^{op} to birhizomic sheaves, with right adjoint π_*, and the question answered here asks whether $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism for every presheaf of abelian groups F on B_P^{op}. We show that it is, for every P, with no finiteness hypothesis. The proof replaces the birhizomic topology by a finer auxiliary topology generated by binary "sign covers" $\{R \to R\langle x\rangle,\ R \to R\langle -x\rangle\}$, where $R\langle \pm x\rangle$ is the universal monoid under R in which $\pm x \in R^{\mathrm{gp}}$ becomes nonnegative; a compactness argument shows that every birhizomic cover is a cover for it, and the left Kan extension of F turns out to be a sheaf for it with split exact Mayer–Vietoris sequences, hence acyclic by an induction on the degree. Acyclicity then descends to the coarser birhizomic topology.

3 Introduction

3.1 Conventions and the objects in the question

Throughout, a *monoid* means a commutative unital monoid M, written additively, that is *integral* (cancellative, so that $M \to M^{\mathrm{gp}}$ is injective, where M^{gp} is the group completion), *saturated* (if $g \in M^{\mathrm{gp}}$ and $ng \in M$ for some integer $n \geq 1$, then $g \in M$), and *sharp* (the only unit is 0). All monoids, homomorphisms, pushouts and colimits below are taken in this category, which we denote by **Mon**. It is a reflective subcategory of commutative monoids (the reflection integralizes, saturates and sharpens), so it has all colimits; the pushouts we actually need to compute are given explicitly in (4)–(5). Basic facts about integral and saturated monoids, localization and sharpening are collected in Ogus's book [4, Ch. I, §§1.3–1.4]; the short verifications we need are included.

Two operations on submonoids of an abelian group G recur constantly. For a submonoid $C \subset G$, the *saturation* of C in G is $C^{\mathrm{sat}} = \{g \in G : ng \in C \text{ for some } n \geq 1\}$, and the *sharpening* of C is $C^\sharp = C/C^\times$, the quotient by the group of units $C^\times = C \cap (-C)$. If C is saturated in G then $C^\sharp \subset G/C^\times$ is integral, saturated and sharp (if $n(g + C^\times) \in C/C^\times$, then $ng + h \in C$ for some $h \in C^\times$, hence $ng \in C$ and $g \in C$).

A monoid M orders its group completion: for $x, y \in M^{\mathrm{gp}}$ put $x \leq y$ if $y - x \in M$. This is a translation-invariant partial order, and M is called *valuative* if the order is total, i.e. $M \cup (-M) = M^{\mathrm{gp}}$. We note once and for all that M^{gp} is torsion-free for every monoid M: if $ng = 0$ with $n \geq 1$, saturation puts g and $-g$ in M, and sharpness forces $g = 0$.

Fix a monoid P; it is arbitrary (in particular, no finite generation is assumed), and it stays fixed throughout. Let A_P be the category of monoids under P, i.e. of homomorphisms $P \to Q$; we often abbreviate an object to Q, and we say that $a \in P$ *dies* in Q, or is *killed* by Q, if its image in Q is 0. For $x \in P$, the *localization* $P[-x] = P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is saturated in P^{gp} (if $mg = p - nx$ with $p \in P$, then $m(g + nx) = p + (m-1)nx \in P$, so $g + nx \in P$ by saturation of P and $g = (g + nx) - nx \in P[-x]$), and

$$L_x := P[-x]^\sharp$$

is the initial monoid under P in which x dies: a homomorphism $P \to T$ sending x to 0 extends uniquely to $P[-x]$ and then kills the units of $P[-x]$ because T is sharp. Let $B_P \subset A_P$ be the full

subcategory with objects $P \to L_x$, $x \in P$. Note $L_0 = P$.

A *presheaf* on the category A_P^{op} is a functor $(A_P^{\mathrm{op}})^{\mathrm{op}} = A_P \to \mathbf{Set}$ (or $\to \mathrm{Ab}$), hence a *covariant* functor on monoids under P; one should think of A_P^{op} as a category of affine "monoid schemes" $\operatorname{Spec} Q$. Accordingly, a family of homomorphisms $\{R \to Q_i\}$ in A_P is the same as a family of maps $\{Q_i \to R\}$ in A_P^{op}, and we refer to it as a family *over* R; fibre products in A_P^{op} over R are pushouts in A_P under R. As usual for such large sites we work in sufficiently large fixed universes, so that sheafification and injective resolutions are available.

3.2 The birhizomic topology

We recall the definitions from the question. For a monoid M, $x \in M^{\mathrm{gp}}$ and $y \in M$, write $x \prec y$ if there are integers m, n with $my \le x \le ny$. (This boundedness notation also appears in Molcho–Wise [2, Def. 2.1.3.1].) A homomorphism $f : M \to N$ is *infinitesimal* if, whenever $x \in M^{\mathrm{gp}}$ and $y \in M$ satisfy $f(x) \in N$, $x \prec y$ and $f(y) \prec f(x)$, one has $x \in M$. If $R \xrightarrow{f} Q \xrightarrow{g} N$ are homomorphisms, f is *birhizomic at* g if f is of finite presentation in **Mon** (discussed in §4) and for every factorization $gf = ph$ with $R \xrightarrow{h} M \xrightarrow{p} N$ and p infinitesimal, there is a unique $u : Q \to M$ with $pu = g$ and $uf = h$. A family $\{R \to Q_i\}_{i \in I}$ is a *birhizomic cover* if for every valuative monoid V and every homomorphism $v : R \to V$ there is a factorization of v through some $Q_i \to V$ such that $R \to Q_i$ is birhizomic at $Q_i \to V$. The question stipulates that the birhizomic covers (of objects R of A_P, all maps being under P) form a Grothendieck topology on A_P^{op}, the *birhizomic topology* J_{b}: a sieve on R is J_{b}-covering when it contains a birhizomic cover. (If one prefers to read J_{b} as the topology *generated* by the birhizomic covers, nothing below changes: Corollary 5.4 exhibits a topology τ for which every birhizomic cover is covering, and τ then contains J_{b} under either reading.) These definitions are taken from the question as stated; a related but different notion of "infinitesimal" homomorphism occurs in [2, Def. 2.1.2.9], and we use only the definition above.

Only two features of this definition enter the proof: (B1) a map birhizomic at anything is of finite presentation; and (B2) in a birhizomic cover of R, every homomorphism from R to a valuative monoid factors through a member of the family that is of finite presentation over R. The infinitesimal lifting condition is never analysed; see Remark 8.1.

3.3 The functors π^{-1} and π_*, and the main theorem

Give B_P^{op} the trivial topology, for which every presheaf is a sheaf, so $\mathrm{Sh}(B_P^{\mathrm{op}}) = \mathrm{PSh}(B_P^{\mathrm{op}})$. Let $j : B_P \hookrightarrow A_P$ be the inclusion. As in the question, $\pi^{-1} : \mathrm{Sh}(B_P^{\mathrm{op}}) \to \mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$ is the composite of the left Kan extension along j with J_{b}-sheafification, and π_* is its right adjoint. Concretely, for a presheaf F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, the left Kan extension is the presheaf

$$(\mathrm{Lan}_j F)(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a), \tag{1}$$

the colimit over the comma category B_P/Q whose objects are pairs (L_a, $L_a \to Q$ under P) [5, Tags 00VC, 00VE]; and π_* is restriction along j: for a sheaf E on A_P^{op}, $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}_{\mathrm{PSh}}(\mathrm{Lan}_j F, E) = \mathrm{Hom}(F, E \circ j)$, so $\pi_* E = E \circ j$. The same formulas define π^{-1} and π_* on abelian (pre)sheaves [5, Tag 00VF]; we check in §6 that the colimit (1) is filtered, so that the Kan extension of an abelian presheaf computed in Ab agrees with the one computed in sets. Since π_*

is left exact on abelian sheaves, it has a right derived functor $\mathrm{R}\pi_*$, and the unit of the adjunction gives a natural map $F \to \pi_*\pi^{-1}F \to \mathrm{R}\pi_*\pi^{-1}F$.

Theorem 3.1. *Let P be an arbitrary monoid (integral, saturated and sharp, as always). For every presheaf of abelian groups F on B_P^{op}, the unit*

$$F \longrightarrow \mathrm{R}\pi_*\pi^{-1}F$$

is a quasi-isomorphism of complexes of abelian presheaves on B_P^{op}, i.e. an isomorphism in the derived category of the category $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ of abelian presheaves. More precisely, the left Kan extension $\mathcal{G} = \mathrm{Lan}_j F$ of (1) is already a J_{b}-sheaf, so $\pi^{-1}F = \mathcal{G}$; the unit $F \to \pi_*\mathcal{G}$ is an isomorphism; and $H^n_{J_{\mathrm{b}}}(Q, \mathcal{G}) = 0$ for every $n > 0$ and every object Q of A_P, where $H^n_{J_{\mathrm{b}}}(Q, -)$ denotes the nth right derived functor of evaluation at Q on abelian J_{b}-sheaves (§7).

The title refers to the following reformulation: since π^{-1} is exact (the colimit (1) is filtered and sheafification is exact), it is left adjoint to $\mathrm{R}\pi_*$ between the bounded below derived categories of $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ and of $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$. Taking $E = \pi^{-1}F'[n]$ in $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}(F, \mathrm{R}\pi_*E)$ and using $F' \simeq \mathrm{R}\pi_*\pi^{-1}F'$, the theorem gives that the map $\mathrm{Ext}^n(F, F') \to \mathrm{Ext}^n(\pi^{-1}F, \pi^{-1}F')$ induced by π^{-1} (which, by naturality of the unit, is the composite of these two identifications) is bijective for all F, F' and n, the Ext groups being computed in the two categories just named.

3.4 Strategy

Cohomology for the birhizomic topology is awkward to compute directly, since its covers are defined by a lifting property. We therefore work with an auxiliary topology τ on A_P^{op} that has an explicit basis and is *finer* than J_{b}. Its generating covers are the *sign covers*

$$\{\, R \to R\langle x\rangle, \quad R \to R\langle -x\rangle \,\}, \qquad x \in R^{\mathrm{gp}},$$

where $R\langle x\rangle$ is the universal monoid under R in which x becomes ≥ 0 (§4). Geometrically, if $x = a - b$ with $a, b \in R$, these are the two charts of the blow-up of the ideal (a, b), saturated and sharpened; this chart construction is classical in toric and logarithmic geometry, see e.g. Nizioł [3, §4]. After the preliminaries on $R\langle x\rangle$ in §4, the proof has four steps.

1. *Comparison of topologies* (§5). Every birhizomic cover is a τ-cover (Corollary 5.4). By (B2) this reduces to Proposition 5.3: a family of finitely presented maps $R \to Q_i$ through which every valuative map $R \to V$ factors is refined by finitely many iterated sign covers. Since R may be infinitely generated, there is no a priori bound on the number of sign choices needed; we obtain one by a compactness argument on the space of "total precones", the saturated submonoids $D \subset R^{\mathrm{gp}}$ with $R \subset D$ and $D \cup (-D) = R^{\mathrm{gp}}$, a closed subset of $\{0, 1\}^{R^{\mathrm{gp}}}$.

2. *The Kan extension is a τ-sheaf with split Mayer–Vietoris sequences* (§6). The value $\mathcal{G}(Q)$ depends only on which elements of P die in Q, so $\mathcal{G}$ is insensitive to *zero-reflecting* maps, i.e. homomorphisms f with $f^{-1}(0) = \{0\}$.

   > **Reviewer 2**
   >
   > usually called sharp or local

   Write $R_\pm = R\langle \pm x\rangle$ for the two members of a sign cover and $R_0 = R\langle x, -x\rangle$ for their "intersection" $R_+ \amalg_R R_-$. The maps from R to R_+, R_- and R_0 are zero-reflecting unless x or

$-x$ already lies in R, in which case the cover contains the identity. In both cases the Mayer–Vietoris sequence $0 \to \mathcal{G}(R) \to \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \to \mathcal{G}(R_0) \to 0$ is split exact (Lemma 6.2).

3. *Acyclicity for τ* (§7). Let $H^n_\tau(R, -)$ denote the nth right derived functor of evaluation at R on abelian τ-sheaves. For a τ-sheaf $\mathcal{E}$ whose sign-cover Mayer–Vietoris sequences are exact on the right, we show $H^n_\tau(R, \mathcal{E}) = 0$ for all $n > 0$ and all R (Lemma 7.1) by induction on n: a class vanishing on both members of a sign cover vanishes (Mayer–Vietoris); every class vanishes on some τ-cover (locality of cohomology); and τ-covers are refined by trees of sign covers, along which the vanishing propagates from the leaves to the root.

4. *Descent to the coarser topology and conclusion* (§7–8). Since τ-sheafification is exact, the inclusion of τ-sheaves into J_{b}-sheaves preserves injectives. A τ-injective resolution of $\mathcal{G}$ is exact on every object by step 3, hence is also a J_{b}-injective resolution, so $H^n_{J_{\mathrm{b}}}(Q, \mathcal{G}) = 0$ for $n > 0$. Since π_* is restriction and $\mathcal{G}$ restricts to F, this is Theorem 3.1.

4 Adjoining inequalities

The basic construction is the universal way of forcing finitely many elements of R^{gp} to become nonnegative.

Definition 4.1. Let R be a monoid, $G = R^{\mathrm{gp}}$, and $S \subset G$ a subset. Put

$$R\langle S\rangle = \Big(\big(R + \textstyle\sum_{s\in S} \mathbb{N}s\big)^{\mathrm{sat}}\Big)^{\sharp}, \tag{2}$$

where the saturation is taken inside G and the sharpening is as in §3.1; thus if $C = (R+\sum_{s\in S}\mathbb{N}s)^{\mathrm{sat}}$, then $R\langle S\rangle = C/(C \cap (-C))$, a monoid with group completion $G/(C \cap (-C))$. It comes with the homomorphism $R \to R\langle S\rangle$ induced by $R \subset C$. For finite sets we write $R\langle s_1, \dots, s_k\rangle$.

Lemma 4.2 (Universal property)**.** *For every monoid T, composition with $R \to R\langle S\rangle$ identifies*

$$\mathrm{Hom}(R\langle S\rangle, T) = \{\, a : R \longrightarrow T : a^{\mathrm{gp}}(s) \in T \text{ for every } s \in S \,\}, \tag{3}$$

where $a^{\mathrm{gp}} : G \to T^{\mathrm{gp}}$ is the induced homomorphism of groups. In particular $R \to R\langle S\rangle$ is an epimorphism in **Mon***, and if S is finite it is of finite presentation.*

> **Reviewer 2**
>
> *I don't think it's of finite presentation, because we're sharpening here. Take a face $F \preceq R$ which is not finitely generated as a monoid but generated by a single element as a face, then adjoint the inverse of that element. Unclear what is meant by fp here – in what category? In* sharp *int sat monoids, I believe it, so I'll assume that's what they mean*

Proof. Let $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$. Then a^{gp} maps $R + \sum_{s\in S}\mathbb{N}s$ into T; it maps C into T because T is saturated (if $ng \in R + \sum \mathbb{N}s$ then $na^{\mathrm{gp}}(g) \in T$, so $a^{\mathrm{gp}}(g) \in T$); and it kills $C \cap (-C)$ because T is sharp. Hence a^{gp} induces $C/(C \cap (-C)) \to T$ extending a. The extension is unique, because a homomorphism out of the integral monoid $R\langle S\rangle$ is determined by the induced map on group completions, and $G \to G/(C \cap (-C))$ is surjective. Conversely, for any $R\langle S\rangle \to T$ the composite a with $R \to R\langle S\rangle$ has $a^{\mathrm{gp}}(s) \in T$, since s maps into $R\langle S\rangle$. Uniqueness of the extension says exactly that $R \to R\langle S\rangle$ is an epimorphism.

For finite presentation, write each $s \in S$ as $s = p_s - q_s$ with $p_s, q_s \in R$. Then (3) says that $R\langle S\rangle$ is obtained from R by adjoining one generator t_s and imposing the relation $q_s + t_s = p_s$, for each $s \in S$, in **Mon**: a homomorphism $R\langle S\rangle \to T$ is a homomorphism $a : R \to T$ together with elements $t_s \in T$ satisfying $a(q_s) + t_s = a(p_s)$, the $t_s = a^{\mathrm{gp}}(s)$ being unique by integrality. In other words, $R\langle S\rangle$ is the quotient, in **Mon**, of $R \oplus \mathbb{N}^S$ by the finitely many relations $q_s + t_s = p_s$. □

The question speaks of finite presentation "in the category of (integral, saturated, sharp) monoids".

> **Reviewer 2**
>
> Ah good, this answers my previous question

One may read this as a presentation by finitely many generators and relations in **Mon** (Q is the quotient in **Mon** of $R \oplus \mathbb{N}^n$ by m relations $u_i = v_i$ with $u_i, v_i \in R \oplus \mathbb{N}^n$, so that $\mathrm{Hom}_R(Q, T)$, the set of homomorphisms under R, is the set of $t \in T^n$ satisfying the m equations obtained by evaluating u_i, v_i at $a : R \to T$ and t), or categorically. Under either reading, the property we shall use is the usual one: *for Q of finite presentation over R, the functor $\mathrm{Hom}_R(Q, -)$ on monoids under R commutes with filtered colimits.* The categorical reading is this property by definition;

> **Reviewer 2**
>
> I don't know what the previous clause means

for the presentation reading it follows from the next lemma, because $\mathrm{Hom}_R(Q, T)$ is then the equalizer of two maps $T^n \rightrightarrows T^m$ natural in T, a finite limit of functors $T \mapsto T^k$, which commute with filtered colimits computed on underlying sets.

> **Reviewer 2**
>
> I don't understand this second statement but I think it's just for the reader's benefit

Lemma 4.3. *Filtered colimits exist in* **Mon** *and are computed as filtered colimits of ordinary commutative monoids, i.e. on underlying sets.*

Proof. Let $T = \varinjlim T_\alpha$ be a filtered colimit of ordinary commutative monoids, with T_α integral, saturated and sharp. Every element of T comes from some stage and every equality in T holds at some later stage; so cancellation ($u + v = u + w \Rightarrow v = w$) and sharpness ($u + v = 0 \Rightarrow u = v = 0$) pass from the T_α to T. Saturation: if $g = u - v \in T^{\mathrm{gp}}$ with $u, v \in T$ and $ng \in T$, say $nu = nv + w$ with $w \in T$, lift u, v, w and this equality to a stage T_α; there $n(u - v) = w \in T_\alpha$, so $u - v \in T_\alpha$ by saturation, and its image in T is g. Thus T lies in **Mon**, and it is then the colimit there. (Integrality and saturation are also covered by [4, Prop. I.1.3.6].) □

Two formal consequences of (3), both proved by comparing the functors $\mathrm{Hom}(-, T)$, are used repeatedly: for subsets $S, S' \subset G$ and a homomorphism $u : R \to R'$,

$$R\langle S\rangle \amalg_R R\langle S'\rangle \simeq R\langle S \cup S'\rangle, \tag{4}$$

$$R' \amalg_R R\langle S\rangle \simeq R'\langle u^{\mathrm{gp}}(S)\rangle. \tag{5}$$

Indeed, a map out of the left side of (4) to T is a map $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$ and $a^{\mathrm{gp}}(S') \subset T$, and a map out of the left side of (5) is a map $b : R' \to T$ with $(bu)^{\mathrm{gp}}(S) = b^{\mathrm{gp}}(u^{\mathrm{gp}}(S)) \subset T$. In particular $R\langle S\rangle\langle u^{\mathrm{gp}}(S')\rangle \simeq R\langle S \cup S'\rangle$ for $u : R \to R\langle S\rangle$.

The remaining two lemmas of this section are the algebraic heart of the argument: the first produces valuative monoids, the second says that adjoining a sign is harmless from the point of view of which elements become zero.

Lemma 4.4 (Extension of an order)**.** *Every monoid R embeds into a valuative monoid W with $W^{\mathrm{gp}} = R^{\mathrm{gp}}$.*

This is [2, Lemma 2.1.2.11]; we include the short proof.

Proof. Let $G = R^{\mathrm{gp}}$. Saturated sharp submonoids of G containing R form a poset under inclusion in which every chain has an upper bound, its union (saturation and sharpness are conditions on finitely many elements at a time). By Zorn's lemma choose a maximal one, C. We claim $C \cup (-C) = G$, so that $W = C$ is valuative with $W^{\mathrm{gp}} = G$.

Suppose that neither x nor $-x$ belongs to C. We show that $C' = (C + \mathbb{N}x)^{\mathrm{sat}}$ is still sharp; since C' is saturated, contains R, and contains $x \notin C$, this contradicts the maximality of C. Let $u, -u \in C'$. Then there are $m, n > 0$, $a, b \in C$ and $i, j \geq 0$ such that

$$mu = a + ix, \qquad -nu = b + jx.$$

Multiplying the first equation by n, the second by m, and adding,

$$0 = na + mb + (ni + mj)x.$$

If $ni+mj > 0$, then $-(ni+mj)x = na+mb \in C$, and saturation of C gives $-x \in C$, a contradiction. Otherwise $i = j = 0$, so mu and $-nu$ lie in C; then mnu and $-mnu$ lie in C, sharpness of C gives $mnu = 0$, and torsion-freeness of G gives $u = 0$. □

Call a homomorphism of monoids $f : Q \to Q'$ *zero-reflecting* if $f^{-1}(0) = \{0\}$. (Between sharp monoids this is the condition that f reflects units, called *local* in [4, p. 15].)

Lemma 4.5. *Let $x \in R^{\mathrm{gp}}$, and set*

$$R_+ = R\langle x\rangle, \qquad R_- = R\langle -x\rangle, \qquad R_0 = R\langle x, -x\rangle.$$

1. *If neither x nor $-x$ lies in R, the three maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting.*
2. *If $x \in R$, then $R_+ = R$, and $R_0 = R_-$ with $R_- \to R_0$ the identity and $R_+ \to R_0$ the map $R \to R_-$. Symmetrically, if $-x \in R$, then $R_- = R$ and $R_0 = R_+$.*

Proof. (2) If $x \in R$, then $R + \mathbb{N}x = R$, which is saturated and sharp, so $R_+ = R$; and $R\langle x, -x\rangle = R\langle -x\rangle$ by (2), since $\mathbb{N}x \subset R$. The description of the maps follows from the uniqueness in Lemma 4.2. The case $-x \in R$ is obtained by replacing x with $-x$.

(1) Since $R \to R_0$ factors through $R \to R_+$ and through $R \to R_-$ (Lemma 4.2), it suffices to treat R_0. Before sharpening, R_0 is $C = (R + \mathbb{Z}x)^{\mathrm{sat}}$. If $r \in R$ maps to zero in $C^\sharp$, then $-r \in C$, so

$$-nr = a + kx \qquad (n > 0,\ a \in R,\ k \in \mathbb{Z}).$$

If $k > 0$, then $-kx = a + nr \in R$, and saturation gives $-x \in R$; if $k < 0$, then $(-k)x = a + nr \in R$ gives $x \in R$. Both are excluded. If $k = 0$, then nr and $-nr$ lie in R, so $nr = 0$ by sharpness and $r = 0$ by torsion-freeness. □

5 A topology finer than the birhizomic topology

Definition 5.1. For an object R of A_P and $x \in R^{\mathrm{gp}}$, the *sign cover* of R at x is the family

$$\{\, R \longrightarrow R\langle x\rangle, \quad R \longrightarrow R\langle -x\rangle \,\} \qquad \text{(written in the monoid direction).} \tag{6}$$

Let τ be the Grothendieck topology on A_P^{op} generated by the sign covers, i.e. the smallest topology for which the sieve generated by every sign cover is covering.

By (5), the base change of the sign cover at x along $u : R \to R'$ (a fibre product in A_P^{op}) is the sign cover of R' at $u^{\mathrm{gp}}(x)$: *sign covers are stable under base change*. This gives τ an explicit description. A *tree family* over R is the family of leaves $\{R \to T_\ell\}_{\ell\in\Lambda}$ of a finite binary tree with root R in which the two children of each internal vertex M form a sign cover of M

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> what does it mean to form a sign cover of a vertex of a tree? It seems that what is meant here is just an iterated sign cover

($R \to T_\ell$ being the composite along the path from the root); its *depth* is the depth of the tree, the trivial tree (depth 0) giving $\{\mathrm{id}_R\}$. Base change along $u : R \to R'$ takes a tree family over R to a tree family over R' of the same depth (apply (5) vertex by vertex), and replacing each leaf of a tree family by a tree family over that leaf (grafting) gives again a tree family. Consequently the sieves containing a tree family form a Grothendieck topology (the three axioms—maximal sieve, stability under base change, local character, i.e. transitivity—follow from the trivial tree, base change and grafting respectively), in which sign covers are covering; conversely, in any topology in which sign covers are covering, every tree family generates a covering sieve, by induction on the depth using the local character axiom. Hence

$$\text{a sieve on } R \text{ is } \tau\text{-covering if and only if it contains a tree family over } R. \tag{7}$$

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> the above seemingly should have been a lemma

For τ, or any other topology J on A_P^{op}, we say that a family over R is a *J-cover* if the sieve it generates is J-covering. Thus a family over R is a τ-cover as soon as it is *refined* by a tree family, meaning that every leaf T_ℓ receives a map under R from some member of the family.

Members of tree families are epimorphisms, being composites of the epimorphisms $M \to M\langle \pm x\rangle$ (Lemma 4.2). Recall that a presheaf $\mathcal{F}$ on A_P^{op} satisfies the *sheaf condition* for a family $\{R \to T_\ell\}_{\ell\in\Lambda}$ if $\mathcal{F}(R) \to \prod_\ell \mathcal{F}(T_\ell)$ is injective with image the *compatible* families (s_ℓ), those for which s_ℓ and s_m have the same image in $\mathcal{F}(T_\ell \amalg_R T_m)$ for all ℓ, m (fibre products in A_P^{op} being pushouts in A_P); when $R \to T_\ell$ is an epimorphism, $T_\ell \amalg_R T_\ell = T_\ell$ and the condition for $\ell = m$ is vacuous. For a sign cover, compatibility of (s_+, s_-) means that s_+ and s_- have the same image in $\mathcal{F}(R\langle x\rangle \amalg_R R\langle -x\rangle) = \mathcal{F}(R\langle x, -x\rangle)$ (4).

Lemma 5.2 (Sheaf criterion)**.** *A presheaf $\mathcal{F}$ of sets or of abelian groups on A_P^{op} is a τ-sheaf as soon as it satisfies the sheaf condition for every sign cover* (6)*.*

Proof. Step A: $\mathcal{F}$ satisfies the sheaf condition for every tree family, over every object. We induct on the depth d; depth 0 is trivial and depth 1 is the hypothesis. Let $\{R \to T_\ell\}$ have depth $d+1 \geq 2$,

let $R_\pm$ be the children of the root and $\Lambda = \Lambda_+ \sqcup \Lambda_-$ the leaves of the two subtrees, which are tree families over $R_\pm$ of depth $\le d$. Let (s_ℓ) be compatible. For $\ell, m \in \Lambda_+$, restricting along the canonical map $T_\ell \amalg_R T_m \to T_\ell \amalg_{R_+} T_m$

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> an isomorphism

shows that $(s_\ell)_{\ell\in\Lambda_+}$ is compatible over R_+, so by induction there is a unique $s_+ \in \mathcal{F}(R_+)$ restricting to s_ℓ for all $\ell \in \Lambda_+$; similarly s_-. We claim that s_+ and s_- have the same image in $\mathcal{F}(R_0)$, $R_0 = R_+ \amalg_R R_-$. The base change of the R_+-subtree along $R_+ \to R_0$ is a tree family over R_0 of depth $\le d$ with leaves $T_\ell \amalg_{R_+} R_0 = T_\ell \amalg_R R_-$ ($\ell \in \Lambda_+$), so by the injectivity part of the induction hypothesis it suffices to compare the restrictions of s_+ and s_- to each $T_\ell \amalg_R R_-$; these are the restrictions of s_ℓ and of s_-. In turn, the base change of the R_--subtree along $R_- \to T_\ell \amalg_R R_-$ is a tree family of depth $\le d$ with leaves $T_\ell \amalg_R T_m$ ($m \in \Lambda_-$), on which s_ℓ and s_- restrict to the images of s_ℓ and s_m, which agree by compatibility; injectivity once more proves the claim. By the sheaf condition for the sign cover $\{R \to R_\pm\}$ there is a unique $s \in \mathcal{F}(R)$ restricting to $s_\pm$, hence to all s_ℓ. If $s' \in \mathcal{F}(R)$ also restricts to all s_ℓ, then $s'|_{R_\pm} = s_\pm$ by the uniqueness in the induction hypothesis, so $s' = s$.

Step B: $\mathcal{F}$ is a τ-sheaf (this is the usual comparison between sheaves for a site and for its associated topology, cf. [5, Tag 00ZC]; we include the argument since tree families, not being closed under isomorphism, do not form a site in the sense of that tag).

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> Very subtle! And unnecessary if isomorphisms are included in the def of sign topology

Let Σ be a τ-covering sieve on R; by (7) it contains a tree family $\{R \to T_\ell\}$. A compatible family on Σ is a collection of sections $s_V \in \mathcal{F}(V)$, one for each arrow $(R \to V) \in \Sigma$ (sections are indexed by arrows; we write s_V when the arrow is clear), with $s_{V'} = s_V|_{V'}$ for every $V \to V'$ under R, and we must show that $s \mapsto (s|_V)_V$ is a bijection from $\mathcal{F}(R)$ onto such collections. Injectivity follows from Step A for $\{R \to T_\ell\}$. Given (s_V), the sections s_{T_ℓ} are compatible (both s_{T_ℓ} and s_{T_m} restrict to $s_{T_\ell \amalg_R T_m}$), so Step A gives $s \in \mathcal{F}(R)$ with $s|_{T_\ell} = s_{T_\ell}$ for all ℓ. For $(R \to V) \in \Sigma$, the sections $s|_V$ and s_V both restrict to $s_{V \amalg_R T_\ell}$ on each leaf of the base-changed tree family $\{V \to V \amalg_R T_\ell\}$, hence are equal by Step A. □

Our goal in this section is to show that τ is finer than J_{b}. The next proposition concerns finitely presented families that "see" all valuative monoids; by (B2), it applies to birhizomic covers. It contains the compactness argument needed for possibly infinitely generated R.

Proposition 5.3. *Let R be an object of A_P and let $\{R \to Q_i\}_{i\in I}$ be a family of maps of finite presentation such that every homomorphism $R \to V$ to a valuative monoid V factors through some $R \to Q_i$. Then the family is refined by a tree family; in particular it is a τ-cover.*

Proof. Step 1: a compact space of orders. Let $G = R^{\mathrm{gp}}$, and let $\mathcal{V}(R)$ be the set of saturated submonoids $D \subset G$ with

$$R \subset D, \qquad D \cup (-D) = G.$$

Such a D is a "total precone": it induces a total preorder on G, but it need not be sharp, and it is essential to allow this: the precones D_ϵ produced in Step 5 below contain the kernel of a surjection $G \to T_\epsilon^{\mathrm{gp}}$ and are not sharp in general. Identify D with its characteristic function and regard

$\mathcal{V}(R)$ as a subset of the product $\{0,1\}^G$, compact by Tychonoff's theorem. It is closed: each of the conditions $r \in D$ (for $r \in R$), $0 \in D$, "$g, h \in D \Rightarrow g + h \in D$", "$ng \in D \Rightarrow g \in D$" and "$g \in D$ or $-g \in D$" involves finitely many coordinates, and $\mathcal{V}(R)$ is the intersection of these closed conditions. Hence $\mathcal{V}(R)$ is compact. (The same device, embedding a set of valuation-like objects into a Boolean product, appears in [1, Lemma 5.3.6].) For finite $S \subset G$, the subset

$$U_S = \{D \in \mathcal{V}(R) : S \subset D\}$$

is open and closed, being defined by finitely many coordinates.

Step 2: the valuative monoid of a precone as a filtered colimit. For $D \in \mathcal{V}(R)$, write $H_D = D \cap (-D)$ and $V_D = D/H_D = D^\sharp$. Then V_D is a monoid (integral, saturated, sharp, cf. §3.1), valuative because $D \cup (-D) = G$, and a monoid under R via $R \subset D \to V_D$. We claim that, in the category of monoids under R,

$$V_D = \varinjlim_{\substack{S \subset D \\ S \text{ finite}}} R\langle S \rangle, \tag{8}$$

the colimit being over the directed set of finite subsets of D, with transition maps $R\langle S\rangle \to R\langle S'\rangle$ for $S \subset S'$ given by Lemma 4.2. Indeed, by (3) a compatible family of maps $R\langle S\rangle \to T$ under $a : R \to T$ is the same as the condition $a^{\mathrm{gp}}(D) \subset T$. Given this condition, $a^{\mathrm{gp}}|_D : D \to T$ kills H_D (as T is sharp) and so factors uniquely through V_D; conversely a map $V_D \to T$ under R composed with $D \to V_D$ shows $a^{\mathrm{gp}}(D) \subset T$. So V_D represents the functor $\varprojlim_S \mathrm{Hom}_R(R\langle S\rangle, -)$, which is the claim.

Step 3: finite presentation yields finite stages. Fix $D \in \mathcal{V}(R)$. By hypothesis the valuative map $R \to V_D$ factors as $R \to Q_i \to V_D$ for some i. Since (8) is a filtered colimit of monoids under R and $\mathrm{Hom}_R(Q_i, -)$ commutes with such colimits (finite presentation; see the discussion following Lemma 4.2), the map $Q_i \to V_D$ factors as

$$Q_i \longrightarrow R\langle S_D \rangle \longrightarrow V_D$$

under R, for some finite $S_D \subset D$. The clopen sets U_{S_D}, $D \in \mathcal{V}(R)$, cover $\mathcal{V}(R)$, since $D \in U_{S_D}$. By compactness choose $D_1, \ldots, D_m$ with $\mathcal{V}(R) = U_{S_1} \cup \cdots \cup U_{S_m}$, where $S_j = S_{D_j}$, and retain the maps

$$Q_{i_j} \longrightarrow R\langle S_j \rangle \qquad (j = 1, \ldots, m) \tag{9}$$

under R. Put $S = S_1 \cup \cdots \cup S_m$, a finite subset of G.

Step 4: the sign tree. Enumerate $S = \{s_1, \ldots, s_k\}$ and form the binary tree of depth k whose vertex at level ℓ with sign history $\epsilon(s_1), \ldots, \epsilon(s_\ell)$ is $R\langle \epsilon(s_1)s_1, \ldots, \epsilon(s_\ell)s_\ell\rangle$; by (5) and (4), the sign cover of this vertex at the image of $s_{\ell+1}$ has as members the two vertices at the next level. The leaves are

$$T_\epsilon = R\langle\, \epsilon(s)s : s \in S \,\rangle, \qquad \epsilon : S \longrightarrow \{1, -1\}, \tag{10}$$

so $\{R \to T_\epsilon\}_\epsilon$ is a tree family. It remains to show that each $R \to T_\epsilon$ factors through some $R\langle S_j\rangle$; composing with (9) then shows that the tree family refines $\{R \to Q_i\}$.

Step 5: each leaf factors through some $R\langle S_j\rangle$. The idea is to attach to each leaf T_ϵ a precone $D_\epsilon \in \mathcal{V}(R)$ extending the inequalities $\epsilon(s)s \geq 0$ that define T_ϵ; the finite subcover gives $S_j \subset D_\epsilon$ for some j, and we then check that the elements of S_j are nonnegative already in T_ϵ. Fix ϵ. By Lemma 4.4, embed T_ϵ into a valuative monoid W_ϵ with $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$. Let $D_\epsilon \subset G$ be the inverse image of W_ϵ under the group homomorphism $G \to T_\epsilon^{\mathrm{gp}} = W_\epsilon^{\mathrm{gp}}$. Then $D_\epsilon \in \mathcal{V}(R)$: it contains R

(which maps into $T_\epsilon \subset W_\epsilon$), it is saturated (as the inverse image of a saturated monoid under a group homomorphism), and $D_\epsilon \cup (-D_\epsilon) = G$ because $W_\epsilon \cup (-W_\epsilon) = W_\epsilon^{\mathrm{gp}}$. Choose j with $D_\epsilon \in U_{S_j}$, i.e. $S_j \subset D_\epsilon$.

We check that every $s \in S_j$ becomes nonnegative in T_ϵ. Let $\bar{s} \in T_\epsilon^{\mathrm{gp}}$ be the image of s. If $\epsilon(s) = 1$, then $\bar{s} \in T_\epsilon$ by construction of T_ϵ. If $\epsilon(s) = -1$, then $-\bar{s} \in T_\epsilon \subset W_\epsilon$, while $s \in D_\epsilon$ says $\bar{s} \in W_\epsilon$; sharpness of W_ϵ gives $\bar{s} = 0$ in $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$, so again $\bar{s} \in T_\epsilon$. (Here it matters that W_ϵ has the *same* group completion as T_ϵ: a map to an arbitrary valuative monoid would only give $\bar{s} \mapsto 0$ there.) By Lemma 4.2 there is a map $R\langle S_j \rangle \to T_\epsilon$ under R, as required. □

Corollary 5.4. *Every birhizomic covering sieve on A_P^{op} is τ-covering; that is, J_{b} is coarser than τ, $J_{\mathrm{b}} \subset \tau$. Consequently every τ-sheaf is a J_{b}-sheaf.*

Proof. Let $\{R \to Q_i\}_{i \in I}$ be a birhizomic cover and let $I' \subset I$ be the set of indices for which $R \to Q_i$ is of finite presentation. By (B1)

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> should cite a lemma, hyperlink, instead of this

and the definition of a birhizomic cover, every homomorphism from R to a valuative monoid factors through some Q_i with $i \in I'$. Proposition 5.3 applied to $\{R \to Q_i\}_{i \in I'}$ shows that the sieve generated by this subfamily, hence also the sieve generated by the whole family, is τ-covering. Every J_{b}-covering sieve contains a birhizomic cover and is therefore τ-covering. The last assertion follows, since a τ-sheaf satisfies the sheaf condition for every τ-covering sieve, in particular for every J_{b}-covering one. □

6 The Kan extension and its Mayer–Vietoris sequences

Fix a presheaf of abelian groups F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, and let $\mathcal{G} = \mathrm{Lan}_j F$ be its left Kan extension (1) to A_P, a presheaf of abelian groups on A_P^{op}:

$$\mathcal{G}(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a).$$

We first make the indexing category explicit. By the universal property of $L_a = P[-a]^\sharp$, a map $L_a \to Q$ under P exists if and only if the image of a in Q is zero, and it is then unique. In particular B_P is a *thin* category (a preorder): there is at most one map $L_a \to L_b$, and it exists if and only if $a \mapsto 0$ in L_b. So B_P/Q is the preorder of elements $a \in P$ that die in Q (elements giving the same object of B_P being identified), with $a \leq b$

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> usually denoted $\preceq$

when a dies in L_b. It is nonempty, since it contains $L_0 = P$, and directed: if a and b die in Q, so does $a + b$, and $a, b \mapsto 0$ in L_{a+b} because the images of a and b in the sharp monoid L_{a+b} sum to zero. Thus (1) is a filtered colimit, and since filtered colimits of abelian groups are computed on underlying sets, the Kan extension of F in abelian presheaves is the same as in presheaves of sets, as announced in the introduction.

Two consequences will be used.

Lemma 6.1. *1. For $Q = L_a$ in B_P, the comma category B_P/L_a has the terminal object (L_a, id), so the unit of the Kan extension is an isomorphism*

$$F(L_a) \xrightarrow{\sim} \mathcal{G}(L_a). \tag{11}$$

2. If $Q \to Q'$ is a zero-reflecting map under P, then an element $a \in P$ dies in Q if and only if it dies in Q', so the comma categories B_P/Q and B_P/Q' coincide, and the restriction map

$$\mathcal{G}(Q) \xrightarrow{\sim} \mathcal{G}(Q') \tag{12}$$

is an isomorphism (it is the identity of the common colimit).

Proof. (1) Every object $(L_b, L_b \to L_a)$ of B_P/L_a has the given map as its unique map to (L_a, id), by thinness; a colimit over a category with a terminal object is the value at that object, and the unit is the corresponding structure map. (2) An element killed in Q is killed in Q'; conversely, if $a \mapsto 0$ in Q', then the image of a in Q is killed by the zero-reflecting map $Q \to Q'$, so it is 0. The restriction map $\mathcal{G}(Q) \to \mathcal{G}(Q')$ is induced by the identity of the diagrams, hence is the identity. □

In words, $\mathcal{G}(Q)$ depends only on the set $\{a \in P : a \text{ dies in } Q\}$ of elements of P killed by Q (a *face* of P: a submonoid containing a and b whenever it contains $a + b$), and $\mathcal{G}$ is "locally constant" along zero-reflecting maps. Combined with Lemma 4.5, which says that sign covers consist of zero-reflecting maps except in the degenerate case, this gives the sheaf property and more.

Lemma 6.2. *For every sign cover* (6)*, with $R_\pm$ and R_0 as in Lemma 4.5, the sequence*

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \longrightarrow \mathcal{G}(R_0) \longrightarrow 0, \tag{13}$$

whose first map is the pair of restrictions and whose second map is the difference of the two restrictions to R_0, is exact, and in fact split. Moreover $\mathcal{G}$ is a τ-sheaf, hence (Corollary 5.4) a J_{b}-sheaf.

Proof. Suppose first that neither x nor $-x$ lies in R. By Lemma 4.5(1) the maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting, so by (12) the three restriction maps out of $\mathcal{G}(R)$ are isomorphisms; since $\mathcal{G}(R) \to \mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ composes to the isomorphism $\mathcal{G}(R) \to \mathcal{G}(R_0)$, the maps $\mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ are isomorphisms too. Identifying all four groups with $A = \mathcal{G}(R)$ through these isomorphisms, every restriction map becomes the identity of A and the sequence becomes

$$0 \longrightarrow A \xrightarrow{a \mapsto (a,a)} A \oplus A \xrightarrow{(a,b) \mapsto a-b} A \longrightarrow 0,$$

which is exact, and split by $a \mapsto (a, 0)$.

If $x \in R$, then by Lemma 4.5(2) $R_+ = R$, $R_0 = R_-$, the map $R_- \to R_0$ is the identity and $R_+ \to R_0$ is $R \to R_-$. Writing $A = \mathcal{G}(R)$, $B = \mathcal{G}(R_-)$ and $\rho : A \to B$ for the restriction, the sequence reads

$$0 \longrightarrow A \xrightarrow{a \mapsto (a, \rho(a))} A \oplus B \xrightarrow{(a,b) \mapsto \rho(a) - b} B \longrightarrow 0,$$

which is exact, split by $(a, b) \mapsto a$, for any homomorphism ρ. The case $-x \in R$ is the same with the roles of R_+ and R_- exchanged.

For the sheaf property, by Lemma 5.2 it suffices to check the sheaf condition for sign covers, and as noted before that lemma this condition says that $\mathcal{G}(R)$ is the equalizer of the two restriction maps $\mathcal{G}(R_+) \times \mathcal{G}(R_-) \rightrightarrows \mathcal{G}(R_0)$, which is exactness of (13) at the first two terms. □

7 Acyclicity, and passage to the coarser topology

For a Grothendieck topology J on A_P^{op}, an object R and an abelian J-sheaf $\mathcal{E}$, we write $H^n_J(R,\mathcal{E})$ for the nth right derived functor of $\mathcal{E}\mapsto\mathcal{E}(R)$ on the abelian category $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}},J)$ of abelian J-sheaves, computed by injective resolutions. We shall use two standard facts. First, the Mayer–Vietoris sequence [5, Tag 0EVY] for a cover $\{R_\pm\to R\}$ in A_P^{op} by two monomorphisms ($R\to R_\pm$ being epimorphisms in A_P) with fibre product R_0; we spell out the version we need in the proof below. Second, *locality of cohomology* [5, Tag 03P1]: for $n>0$, every class $\xi\in H^n_J(R,\mathcal{E})$ vanishes on the members of some J-cover of R (in the sense of §5). (Explicitly: if $\mathcal{E}\to\mathcal{I}^\bullet$ is an injective resolution and $z\in\mathcal{I}^n(R)$ a cocycle representing ξ, the exactness of $\mathcal{I}^\bullet$ as a complex of *sheaves* in degree $n>0$ says that z is locally a coboundary.)

Lemma 7.1. *Let $\mathcal{E}$ be an abelian τ-sheaf on A_P^{op} such that for every sign cover* (6)*, with $R_\pm$ and R_0 as in Lemma 4.5, the difference map*

$$\mathcal{E}(R_+)\oplus\mathcal{E}(R_-)\longrightarrow\mathcal{E}(R_0) \tag{14}$$

is surjective. Then $H^n_\tau(R,\mathcal{E})=0$ for all $n>0$ and all objects R of A_P.

Proof. Mayer–Vietoris. For an object Y of A_P let h_Y be the presheaf represented by Y on A_P^{op}, so $h_Y(X)=\mathrm{Hom}_{A_P}(Y,X)$, and let $\mathbb{Z}_\tau[Y]$ be the τ-sheaf associated with the free abelian presheaf $\mathbb{Z}[h_Y]$; then $\mathrm{Hom}(\mathbb{Z}_\tau[Y],\mathcal{E})=\mathcal{E}(Y)$ for every abelian τ-sheaf $\mathcal{E}$ [5, Tag 03AB], so that $H^n_\tau(Y,\mathcal{E})=\mathrm{Ext}^n(\mathbb{Z}_\tau[Y],\mathcal{E})$, the Ext group being computed in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}},\tau)$ (apply $\mathrm{Hom}(\mathbb{Z}_\tau[Y],-)$ to an injective resolution of $\mathcal{E}$). Fix a sign cover of R at x.

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> "at x?"

For every X, the universal property (3) identifies $h_{R_\pm}(X)$ with subsets of $h_R(X)$, with intersection $h_{R_0}(X)$ by (4). The elementary exact sequence $0\to\mathbb{Z}[U\cap U']\to\mathbb{Z}[U]\oplus\mathbb{Z}[U']\to\mathbb{Z}[U\cup U']\to 0$ of free abelian groups on two subsets of a set therefore gives an exact sequence of presheaves

$$0\longrightarrow\mathbb{Z}[h_{R_0}]\longrightarrow\mathbb{Z}[h_{R_+}]\oplus\mathbb{Z}[h_{R_-}]\longrightarrow\mathbb{Z}[h_R]$$

whose cokernel is $\mathbb{Z}[h_R]/\mathbb{Z}[h_{R_+}\cup h_{R_-}]$. This cokernel has zero τ-sheafification: a section of h_R over X is a map $f:R\to X$, and on the two members of the sign cover of X at $f^{\mathrm{gp}}(x)$, which are $X\amalg_R R_\pm$ by (5), the restrictions of f lie in h_{R_+} and h_{R_-} respectively. Since sheafification is exact we obtain a short exact sequence of τ-sheaves

$$0\longrightarrow\mathbb{Z}_\tau[R_0]\longrightarrow\mathbb{Z}_\tau[R_+]\oplus\mathbb{Z}_\tau[R_-]\longrightarrow\mathbb{Z}_\tau[R]\longrightarrow 0,$$

and applying $\mathrm{Ext}^\bullet(-,\mathcal{E})$ the long exact sequence

$$\begin{aligned}0\to\mathcal{E}(R)\to\mathcal{E}(R_+)\oplus\mathcal{E}(R_-)\to\mathcal{E}(R_0)\to H^1_\tau(R,\mathcal{E})\to\cdots\\ \cdots\to H^{n-1}_\tau(R_0,\mathcal{E})\to H^n_\tau(R,\mathcal{E})\to H^n_\tau(R_+,\mathcal{E})\oplus H^n_\tau(R_-,\mathcal{E})\to H^n_\tau(R_0,\mathcal{E})\to\cdots\end{aligned} \tag{15}$$

This is the Mayer–Vietoris sequence of [5, Tag 0EVY] for the present situation.

Injectivity on the members of a sign cover. We prove the lemma by induction on $n\geq 1$, simultaneously for all objects R. Consider the statement $(*)_n$ that, for every sign cover of every object R, the map

$$H^n_\tau(R,\mathcal{E})\longrightarrow H^n_\tau(R_+,\mathcal{E})\oplus H^n_\tau(R_-,\mathcal{E})$$

is injective. For $n = 1$, $(*)_1$ follows from (15) and the assumed surjectivity of (14), which makes the connecting map $\mathcal{E}(R_0) \to H^1_\tau(R, \mathcal{E})$ zero. For $n > 1$, if $H^{n-1}_\tau(-, \mathcal{E}) = 0$ on all objects,

> **Reviewer 2**
>
> isn't this what is being shown? Edit: The proof means to induct for the proof of the *whole* lemma, not just this stage of the argument. This was confusing and circular if interpreted in the latter fashion.

then in particular $H^{n-1}_\tau(R_0, \mathcal{E}) = 0$ and (15) gives $(*)_n$.

From the leaves to the root. Having established $(*)_n$ (unconditionally for $n = 1$, and under the induction hypothesis for $n > 1$), let $\xi \in H^n_\tau(R, \mathcal{E})$. By locality of cohomology, ξ vanishes on the members of some τ-cover of R. By (7), the sieve generated by this family contains a tree family $\{R \to T_\ell\}$ (each T_ℓ receives a map under R from a member on which ξ vanishes), so $\xi|_{T_\ell} = 0$ for every leaf. Now argue upward through the tree: if M is an internal vertex with children M_+, M_- forming a sign cover of M, and $\xi|_{M_+} = 0 = \xi|_{M_-}$, then $\xi|_M = 0$ by $(*)_n$ applied to the sign cover of M. Induction on the height of a vertex gives $\xi|_R = \xi = 0$. Hence $H^n_\tau(R, \mathcal{E}) = 0$ for all R, which completes the induction on n. □

Corollary 7.2. *Let $\mathcal{G} = \mathrm{Lan}_j F$ be the Kan extension of §6. For every object R of A_P and every $n > 0$, $H^n_\tau(R, \mathcal{G}) = 0$.*

Proof. By Lemma 6.2, $\mathcal{G}$ is a τ-sheaf whose maps (14) are surjective. Apply Lemma 7.1. □

We now transfer this vanishing to the coarser topology J_{b}. Some care is needed, since the inclusion of τ-sheaves into J_{b}-sheaves is not exact in general (cokernels involve different sheafifications); what does transfer is injectivity of objects, and Corollary 7.2 makes a τ-injective resolution exact already as a complex of presheaves.

Proposition 7.3. *For $\mathcal{G} = \mathrm{Lan}_j F$, every object R of A_P and every $n > 0$, $H^n_{J_{\mathrm{b}}}(R, \mathcal{G}) = 0$.*

Proof. Since $J_{\mathrm{b}} \subset \tau$ (Corollary 5.4), the identity of A_P^{op} is a morphism of sites $(A_P^{\mathrm{op}}, \tau) \to (A_P^{\mathrm{op}}, J_{\mathrm{b}})$ whose direct image is the inclusion

$$\iota : \mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, \tau) \longrightarrow \mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$$

and whose inverse image is τ-sheafification, an exact functor [5, Tag 0EWK]. A right adjoint whose left adjoint is exact preserves injectives [5, Tag 015Z]; hence ι carries injective τ-sheaves to injective J_{b}-sheaves.

Choose an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, \tau)$. By Corollary 7.2, for every object R the augmented complex of abelian groups

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{I}^0(R) \longrightarrow \mathcal{I}^1(R) \longrightarrow \cdots \tag{16}$$

is exact. So $\mathcal{G} \to \mathcal{I}^\bullet$ is exact as a complex of presheaves, hence exact as a complex of J_{b}-sheaves (kernels of maps of sheaves are computed in presheaves, and a presheaf image that is already a sheaf equals the sheaf image). Its terms $\iota\mathcal{I}^k$ are injective J_{b}-sheaves by the preceding paragraph. Thus $\mathcal{G} \to \iota\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{G}$ in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$, and $H^n_{J_{\mathrm{b}}}(R, \mathcal{G})$ is the nth cohomology of the same complex (16), which vanishes for $n > 0$. □

8 Proof of Theorem 3.1

Recall from §3.3 that $\pi^{-1}F$ is the J_{b}-sheafification of the left Kan extension $\mathcal{G} = \mathrm{Lan}_j F$, and that π_* is restriction along $j : B_P \hookrightarrow A_P$.

$\pi^{-1}F = \mathcal{G}$. By Lemma 6.2, $\mathcal{G}$ is already a J_{b}-sheaf, so no sheafification is needed and $\pi^{-1}F = \mathcal{G}$.

The unit in degree zero. The unit $F \to \pi_*\pi^{-1}F = \mathcal{G} \circ j$ evaluated at L_a is the Kan extension unit $F(L_a) \to \mathcal{G}(L_a)$, an isomorphism by (11).

Vanishing of $R^n\pi_*$ for $n > 0$. Let $\mathcal{G} \to \mathcal{I}^\bullet$ be an injective resolution in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$. By definition $\mathrm{R}\pi_*\mathcal{G}$ is represented by the complex of presheaves $\pi_*\mathcal{I}^\bullet = \mathcal{I}^\bullet \circ j$ on B_P^{op}. Since B_P^{op} carries the trivial topology, evaluation at any object L_a is an exact functor on $\mathrm{Ab}\,\mathrm{Sh}(B_P^{\mathrm{op}}) = \mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$, so

$$(R^n\pi_*\mathcal{G})(L_a) = H^n\big(\mathcal{I}^\bullet(L_a)\big) = H^n_{J_{\mathrm{b}}}(L_a, \mathcal{G}) = 0 \qquad (n > 0)$$

by Proposition 7.3. As the L_a are all the objects of B_P, the cohomology presheaves $R^n\pi_*\mathcal{G}$ vanish for $n > 0$, and $R^0\pi_*\mathcal{G} = \pi_*\mathcal{G} \cong F$ by the previous paragraph. Hence $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism. This proves Theorem 3.1, including the three "more precise" assertions. □

Remark 8.1 (Where the hypotheses are used). The infinitesimal lifting condition in the definition of a birhizomic cover is never analysed: Proposition 5.3 shows that *any* family of finitely presented maps through which every valuative map factors is a τ-cover, and this is all that is used. Finite presentation enters exactly once, in Step 3 of that proof, to factor $Q_i \to V_D$ through a finite stage of the colimit (8); the colimit identity itself needs no finiteness. Finite generation of P is never used, which is why the compactness argument on $\mathcal{V}(R)$ is needed in place of a dimension count. Finally, the argument of Proposition 7.3 shows that $\mathcal{G}$ is acyclic for every topology coarser than τ, the birhizomic topology being one instance.

Remark 8.2 (Two small examples). For $R = \mathbb{N}^2$ and $x = e_1 - e_2$, neither sign lies in R; the members of the sign cover are $\mathbb{N}^2\langle\pm(e_1 - e_2)\rangle$, the two charts of the blow-up of the origin, and $R_0 = \mathbb{N}^2\langle x, -x\rangle \simeq \mathbb{N}$ via $(a, b) \mapsto a + b$. None of these maps kills a nonzero element of $\mathbb{N}^2$, so $\mathcal{G}$ takes the same value on all four monoids, as in the first case of Lemma 6.2. For $R = \mathbb{N}$ and $x = 1$ the sign cover is $\{\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to 0\}$ with $R_0 = 0$, and (13) is $0 \to \mathcal{G}(\mathbb{N}) \to \mathcal{G}(\mathbb{N}) \oplus \mathcal{G}(0) \to \mathcal{G}(0) \to 0$. Deeper in a sign tree a leaf may kill nonzero elements of P; this is harmless, as Lemma 4.5 is applied at each internal vertex.

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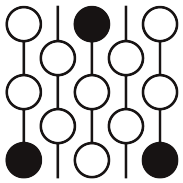
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[5] The Stacks Project Authors, *The Stacks Project*, 2026. Tags 00VC, 00VE, 00VF (Kan extension along a functor and its adjoint), 00ZC (the topology defined by a site), 03AB (free abelian sheaves), 015Z (exact left adjoints and injectives), 03P1 (locality of cohomology), 0EVY (Mayer–Vietoris on a site), 0EWK (comparing two topologies).

Review of Problem 1 – ProofCouncil



First Proof Third Batch
Reviewer 3
October 8–9, 2026

1 Review

Global opinion: This is by a clear way the best of the submissions, and also better than the human solution. It has a good strategy, and solves the problem fully. There is some novelty in the arguments compared to the human solution: in one reduction step, it chooses a slightly smarter and smaller case to reduce too. The reduction step is essentially the same difficult, but the smaller case is easier than what the other models or the human solution need to do.
It sets up more structure than the human solution, which I personally enjoy, and I think makes it easier to follow in this aspect.
It has the best exposition from all models, also better than the human solution. It gives a clear overview of the overall strategy, which is also readable without understanding the specific jargon/notation used by this model. Individual proofs are also well-structured. It also includes helpful remarks, and even an example at the end. Attribution is great.

This feels the most like it could go on arxiv tomorrow as a stand alone paper.

Correctness. It is essentially flawless. I commend it for the reduction step from birhizomic covers to some kind of finite covers. It reduces essentially to a single case (the sign cover), while other models and the human solution reduce to a more general case. The reduction is the same difficulty in either case, but the more general case involves way more algebra to prove.
Score: 1

Exposition. It is well-written and clear. It clearly explains its strategy, best (most human-readable and instantly understandable) of all the models and the human solution. It feels like it has thought a lot about what to say and what not to say, and kept its audience in mind.

It prefers long sentences, and sometimes has proofs in parentheses in these long sentences. I'd ask the author to revise this if I were a referee, but I would say that it is still clear to follow.
Score: 1

Attributions. It is very self-contained. The citations it does make check out, for the rest it's mostly doing rather elementary homological/monoid algebra. It cites more than Lynx, less than Wren (which cites too much); this feels like the golden middle.
Score: 1

2 Submission with inline comments

Abstract

Let P be an integral, saturated, sharp commutative monoid, let A_P be the category of monoids under P, equipped with the birhizomic topology on A_P^{op}, and let $B_P \subset A_P$ be the full subcategory of the sharpened localizations $P \to P[-x]^\sharp$, $x \in P$, with the trivial topology (all definitions are recalled in §1). The inclusion induces an inverse image functor π^{-1} from presheaves on B_P^{op} to birhizomic sheaves, with right adjoint π_*, and the question answered here asks whether $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism for every presheaf of abelian groups F on B_P^{op}. We show that it is, for every P, with no finiteness hypothesis. The proof replaces the birhizomic topology by a finer auxiliary topology generated by binary "sign covers" $\{R \to R\langle x\rangle,\ R \to R\langle -x\rangle\}$, where $R\langle \pm x\rangle$ is the universal monoid under R in which $\pm x \in R^{\mathrm{gp}}$ becomes nonnegative; a compactness argument shows that every birhizomic cover is a cover for it, and the left Kan extension of F turns out to be a sheaf for it with split exact Mayer–Vietoris sequences, hence acyclic by an induction on the degree. Acyclicity then descends to the coarser birhizomic topology.

3 Introduction

3.1 Conventions and the objects in the question

Throughout, a *monoid* means a commutative unital monoid M, written additively, that is *integral* (cancellative, so that $M \to M^{\mathrm{gp}}$ is injective, where M^{gp} is the group completion), *saturated* (if $g \in M^{\mathrm{gp}}$ and $ng \in M$ for some integer $n \geq 1$, then $g \in M$), and *sharp* (the only unit is 0). All monoids, homomorphisms, pushouts and colimits below are taken in this category, which we denote by **Mon**. It is a reflective subcategory of commutative monoids (the reflection integralizes, saturates and sharpens), so it has all colimits; the pushouts we actually need to compute are given explicitly in (4)–(5). Basic facts about integral and saturated monoids, localization and sharpening are collected in Ogus's book [4, Ch. I, §§1.3–1.4]; the short verifications we need are included.

Two operations on submonoids of an abelian group G recur constantly. For a submonoid $C \subset G$, the *saturation* of C in G is $C^{\mathrm{sat}} = \{g \in G : ng \in C \text{ for some } n \geq 1\}$, and the *sharpening* of C is $C^\sharp = C/C^\times$, the quotient by the group of units $C^\times = C \cap (-C)$. If C is saturated in G then $C^\sharp \subset G/C^\times$ is integral, saturated and sharp (if $n(g + C^\times) \in C/C^\times$, then $ng + h \in C$ for some $h \in C^\times$, hence $ng \in C$ and $g \in C$).

A monoid M orders its group completion: for $x, y \in M^{\mathrm{gp}}$ put $x \leq y$ if $y - x \in M$. This is a translation-invariant partial order, and M is called *valuative* if the order is total, i.e. $M \cup (-M) = M^{\mathrm{gp}}$. We note once and for all that M^{gp} is torsion-free for every monoid M: if $ng = 0$ with $n \geq 1$, saturation puts g and $-g$ in M, and sharpness forces $g = 0$.

Fix a monoid P; it is arbitrary (in particular, no finite generation is assumed), and it stays fixed throughout. Let A_P be the category of monoids under P, i.e. of homomorphisms $P \to Q$; we often abbreviate an object to Q, and we say that $a \in P$ *dies* in Q, or is *killed* by Q, if its image in Q is 0. For $x \in P$, the *localization* $P[-x] = P + \mathbb{N}(-x) \subset P^{\mathrm{gp}}$ is saturated in P^{gp} (if $mg = p - nx$ with $p \in P$, then $m(g + nx) = p + (m-1)nx \in P$, so $g + nx \in P$ by saturation of P and $g = (g + nx) - nx \in P[-x]$), and

$$L_x := P[-x]^\sharp$$

is the initial monoid under P in which x dies: a homomorphism $P \to T$ sending x to 0 extends uniquely to $P[-x]$ and then kills the units of $P[-x]$ because T is sharp. Let $B_P \subset A_P$ be the full

subcategory with objects $P \to L_x$, $x \in P$. Note $L_0 = P$.

> **Reviewer 3**
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> Exp: lots of inline in parentheses proofs in above page

A *presheaf* on the category A_P^{op} is a functor $(A_P^{\mathrm{op}})^{\mathrm{op}} = A_P \to \mathbf{Set}$ (or $\to$ Ab), hence a *covariant* functor on monoids under P; one should think of A_P^{op} as a category of affine "monoid schemes" $\operatorname{Spec} Q$. Accordingly, a family of homomorphisms $\{R \to Q_i\}$ in A_P is the same as a family of maps $\{Q_i \to R\}$ in A_P^{op}, and we refer to it as a family *over* R; fibre products in A_P^{op} over R are pushouts in A_P under R. As usual for such large sites we work in sufficiently large fixed universes, so that sheafification and injective resolutions are available.

3.2 The birhizomic topology

We recall the definitions from the question. For a monoid M, $x \in M^{\mathrm{gp}}$ and $y \in M$, write $x \prec y$ if there are integers m, n with $my \le x \le ny$. (This boundedness notation also appears in Molcho–Wise [2, Def. 2.1.3.1].) A homomorphism $f : M \to N$ is *infinitesimal* if, whenever $x \in M^{\mathrm{gp}}$ and $y \in M$ satisfy $f(x) \in N$, $x \prec y$ and $f(y) \prec f(x)$, one has $x \in M$. If $R \xrightarrow{f} Q \xrightarrow{g} N$ are homomorphisms, f is *birhizomic at* g if f is of finite presentation in **Mon** (discussed in §4) and for every factorization $gf = ph$ with $R \xrightarrow{h} M \xrightarrow{p} N$ and p infinitesimal, there is a unique $u : Q \to M$ with $pu = g$ and $uf = h$. A family $\{R \to Q_i\}_{i \in I}$ is a *birhizomic cover* if for every valuative monoid V and every homomorphism $v : R \to V$ there is a factorization of v through some $Q_i \to V$ such that $R \to Q_i$ is birhizomic at $Q_i \to V$. The question stipulates that the birhizomic covers (of objects R of A_P, all maps being under P) form a Grothendieck topology on A_P^{op}, the *birhizomic topology* J_{b}: a sieve on R is J_{b}-covering when it contains a birhizomic cover. (If one prefers to read J_{b} as the topology *generated* by the birhizomic covers, nothing below changes: Corollary 5.4 exhibits a topology τ for which every birhizomic cover is covering, and τ then contains J_{b} under either reading.) These definitions are taken from the question as stated; a related but different notion of "infinitesimal" homomorphism occurs in [2, Def. 2.1.2.9], and we use only the definition above.

Only two features of this definition enter the proof: (B1) a map birhizomic at anything is of finite presentation; and (B2) in a birhizomic cover of R, every homomorphism from R to a valuative monoid factors through a member of the family that is of finite presentation over R. The infinitesimal lifting condition is never analysed; see Remark 8.1.

> **Reviewer 3**
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> Cor: hmm, it claims it doesn't need infinitesimal lifting? I thought there might be a counterxample without (in particular, for base changes of Bl $\mathbb{A}^2$). But maybe that enters only in the scheme case. Ask another referee

3.3 The functors π^{-1} and π_*, and the main theorem

Give B_P^{op} the trivial topology, for which every presheaf is a sheaf, so $\mathrm{Sh}(B_P^{\mathrm{op}}) = \mathrm{PSh}(B_P^{\mathrm{op}})$. Let $j : B_P \hookrightarrow A_P$ be the inclusion. As in the question, $\pi^{-1} : \mathrm{Sh}(B_P^{\mathrm{op}}) \to \mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$ is the composite of the left Kan extension along j with J_{b}-sheafification, and π_* is its right adjoint. Concretely, for

a presheaf F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, the left Kan extension is the presheaf

$$(\mathrm{Lan}_j F)(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a), \tag{1}$$

the colimit over the comma category B_P/Q whose objects are pairs $(L_a,\ L_a \to Q \text{ under } P)$ [5, Tags 00VC, 00VE]; and π_* is restriction along j: for a sheaf E on A_P^{op}, $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}_{\mathrm{PSh}}(\mathrm{Lan}_j F, E) = \mathrm{Hom}(F, E \circ j)$, so $\pi_* E = E \circ j$. The same formulas define π^{-1} and π_* on abelian (pre)sheaves [5, Tag 00VF]; we check in §6 that the colimit (1) is filtered, so that the Kan extension of an abelian presheaf computed in Ab agrees with the one computed in sets. Since π_* is left exact on abelian sheaves, it has a right derived functor $\mathrm{R}\pi_*$, and the unit of the adjunction gives a natural map $F \to \pi_*\pi^{-1}F \to \mathrm{R}\pi_*\pi^{-1}F$.

Theorem 3.1. *Let P be an arbitrary monoid (integral, saturated and sharp, as always). For every presheaf of abelian groups F on B_P^{op}, the unit*

$$F \longrightarrow \mathrm{R}\pi_*\pi^{-1}F$$

is a quasi-isomorphism of complexes of abelian presheaves on B_P^{op}, i.e. an isomorphism in the derived category of the category $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ of abelian presheaves. More precisely, the left Kan extension $\mathcal{G} = \mathrm{Lan}_j F$ of (1) is already a J_{b}-sheaf, so $\pi^{-1}F = \mathcal{G}$; the unit $F \to \pi_*\mathcal{G}$ is an isomorphism; and $H^n_{J_{\mathrm{b}}}(Q, \mathcal{G}) = 0$ for every $n > 0$ and every object Q of A_P, where $H^n_{J_{\mathrm{b}}}(Q, -)$ denotes the nth right derived functor of evaluation at Q on abelian J_{b}-sheaves (§7).

The title refers to the following reformulation: since π^{-1} is exact (the colimit (1) is filtered and sheafification is exact), it is left adjoint to $\mathrm{R}\pi_*$ between the bounded below derived categories of $\mathrm{Ab}\,\mathrm{PSh}(B_P^{\mathrm{op}})$ and of $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J_{\mathrm{b}})$. Taking $E = \pi^{-1}F'[n]$ in $\mathrm{Hom}(\pi^{-1}F, E) = \mathrm{Hom}(F, \mathrm{R}\pi_* E)$ and using $F' \simeq \mathrm{R}\pi_*\pi^{-1}F'$, the theorem gives that the map $\mathrm{Ext}^n(F, F') \to \mathrm{Ext}^n(\pi^{-1}F, \pi^{-1}F')$ induced by π^{-1} (which, by naturality of the unit, is the composite of these two identifications) is bijective for all F, F' and n, the Ext groups being computed in the two categories just named.

3.4 Strategy

> **Reviewer 3**
>
> The following is extremely clear.

Cohomology for the birhizomic topology is awkward to compute directly, since its covers are defined by a lifting property. We therefore work with an auxiliary topology τ on A_P^{op} that has an explicit basis and is *finer* than J_{b}. Its generating covers are the *sign covers*

$$\{\, R \to R\langle x\rangle, \quad R \to R\langle -x\rangle \,\}, \qquad x \in R^{\mathrm{gp}},$$

where $R\langle x\rangle$ is the universal monoid under R in which x becomes ≥ 0 (§4). Geometrically, if $x = a - b$ with $a, b \in R$, these are the two charts of the blow-up of the ideal (a, b), saturated and sharpened; this chart construction is classical in toric and logarithmic geometry, see e.g. Nizioł [3, §4]. After the preliminaries on $R\langle x\rangle$ in §4, the proof has four steps.

1. *Comparison of topologies* (§5). Every birhizomic cover is a τ-cover (Corollary 5.4). By (B2) this reduces to Proposition 5.3: a family of finitely presented maps $R \to Q_i$ through which

every valuative map $R \to V$ factors is refined by finitely many iterated sign covers. Since R may be infinitely generated, there is no a priori bound on the number of sign choices needed; we obtain one by a compactness argument on the space of "total precones", the saturated submonoids $D \subset R^{\mathrm{gp}}$ with $R \subset D$ and $D \cup (-D) = R^{\mathrm{gp}}$, a closed subset of $\{0,1\}^{R^{\mathrm{gp}}}$.

2. *The Kan extension is a τ-sheaf with split Mayer–Vietoris sequences* (§6). The value $\mathcal{G}(Q)$ depends only on which elements of P die in Q, so $\mathcal{G}$ is insensitive to *zero-reflecting* maps, i.e. homomorphisms f with $f^{-1}(0) = \{0\}$. Write $R_\pm = R\langle \pm x\rangle$ for the two members of a sign cover and $R_0 = R\langle x, -x\rangle$ for their "intersection" $R_+ \amalg_R R_-$. The maps from R to R_+, R_- and R_0 are zero-reflecting unless x or $-x$ already lies in R, in which case the cover contains the identity. In both cases the Mayer–Vietoris sequence $0 \to \mathcal{G}(R) \to \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \to \mathcal{G}(R_0) \to 0$ is split exact (Lemma 6.2).

3. *Acyclicity for τ* (§7). Let $H^n_\tau(R, -)$ denote the nth right derived functor of evaluation at R on abelian τ-sheaves. For a τ-sheaf $\mathcal{E}$ whose sign-cover Mayer–Vietoris sequences are exact on the right, we show $H^n_\tau(R, \mathcal{E}) = 0$ for all $n > 0$ and all R (Lemma 7.1) by induction on n: a class vanishing on both members of a sign cover vanishes (Mayer–Vietoris); every class vanishes on some τ-cover (locality of cohomology); and τ-covers are refined by trees of sign covers, along which the vanishing propagates from the leaves to the root.

4. *Descent to the coarser topology and conclusion* (§7–8). Since τ-sheafification is exact, the inclusion of τ-sheaves into J_{b}-sheaves preserves injectives. A τ-injective resolution of $\mathcal{G}$ is exact on every object by step 3, hence is also a J_{b}-injective resolution, so $H^n_{J_{\mathrm{b}}}(Q, \mathcal{G}) = 0$ for $n > 0$. Since π_* is restriction and $\mathcal{G}$ restricts to F, this is Theorem 3.1.

4 Adjoining inequalities

The basic construction is the universal way of forcing finitely many elements of R^{gp} to become nonnegative.

Definition 4.1. Let R be a monoid, $G = R^{\mathrm{gp}}$, and $S \subset G$ a subset. Put

$$R\langle S\rangle = \Big(\big(R + \textstyle\sum_{s\in S} \mathbb{N}s\big)^{\mathrm{sat}}\Big)^\sharp, \tag{2}$$

where the saturation is taken inside G and the sharpening is as in §3.1; thus if $C = (R + \sum_{s\in S} \mathbb{N}s)^{\mathrm{sat}}$, then $R\langle S\rangle = C/(C \cap (-C))$, a monoid with group completion $G/(C \cap (-C))$. It comes with the homomorphism $R \to R\langle S\rangle$ induced by $R \subset C$. For finite sets we write $R\langle s_1, \dots, s_k\rangle$.

Lemma 4.2 (Universal property)**.** *For every monoid T, composition with $R \to R\langle S\rangle$ identifies*

$$\mathrm{Hom}(R\langle S\rangle, T) = \{\, a : R \longrightarrow T : a^{\mathrm{gp}}(s) \in T \text{ for every } s \in S \,\}, \tag{3}$$

where $a^{\mathrm{gp}} : G \to T^{\mathrm{gp}}$ is the induced homomorphism of groups. In particular $R \to R\langle S\rangle$ is an epimorphism in **Mon***, and if S is finite it is of finite presentation.*

Proof. Let $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$. Then a^{gp} maps $R + \sum_{s\in S} \mathbb{N}s$ into T; it maps C into T because T is saturated (if $ng \in R + \sum \mathbb{N}s$ then $na^{\mathrm{gp}}(g) \in T$, so $a^{\mathrm{gp}}(g) \in T$); and it kills $C \cap (-C)$ because T is sharp. Hence a^{gp} induces $C/(C \cap (-C)) \to T$ extending a. The extension is unique,

because a homomorphism out of the integral monoid $R\langle S\rangle$ is determined by the induced map on group completions, and $G \to G/(C \cap (-C))$ is surjective. Conversely, for any $R\langle S\rangle \to T$ the composite a with $R \to R\langle S\rangle$ has $a^{\mathrm{gp}}(s) \in T$, since s maps into $R\langle S\rangle$. Uniqueness of the extension says exactly that $R \to R\langle S\rangle$ is an epimorphism.

For finite presentation, write each $s \in S$ as $s = p_s - q_s$ with $p_s, q_s \in R$. Then (3) says that $R\langle S\rangle$ is obtained from R by adjoining one generator t_s and imposing the relation $q_s + t_s = p_s$, for each $s \in S$, in **Mon**: a homomorphism $R\langle S\rangle \to T$ is a homomorphism $a : R \to T$ together with elements $t_s \in T$ satisfying $a(q_s) + t_s = a(p_s)$, the $t_s = a^{\mathrm{gp}}(s)$ being unique by integrality. In other words, $R\langle S\rangle$ is the quotient, in **Mon**, of $R \oplus \mathbb{N}^S$ by the finitely many relations $q_s + t_s = p_s$. □

The question speaks of finite presentation "in the category of (integral, saturated, sharp) monoids". One may read this as a presentation by finitely many generators and relations in **Mon** (Q is the quotient in **Mon** of $R \oplus \mathbb{N}^n$ by m relations $u_i = v_i$ with $u_i, v_i \in R \oplus \mathbb{N}^n$, so that $\mathrm{Hom}_R(Q, T)$, the set of homomorphisms under R, is the set of $t \in T^n$ satisfying the m equations obtained by evaluating u_i, v_i at $a : R \to T$ and t), or categorically. Under either reading, the property we shall use is the usual one: *for Q of finite presentation over R, the functor $\mathrm{Hom}_R(Q, -)$ on monoids under R commutes with filtered colimits.* The categorical reading is this property by definition; for the presentation reading it follows from the next lemma, because $\mathrm{Hom}_R(Q, T)$ is then the equalizer of two maps $T^n \rightrightarrows T^m$ natural in T, a finite limit of functors $T \mapsto T^k$, which commute with filtered colimits computed on underlying sets.

Lemma 4.3. *Filtered colimits exist in* **Mon** *and are computed as filtered colimits of ordinary commutative monoids, i.e. on underlying sets.*

Proof. Let $T = \varinjlim T_\alpha$ be a filtered colimit of ordinary commutative monoids, with T_α integral, saturated and sharp. Every element of T comes from some stage and every equality in T holds at some later stage; so cancellation ($u + v = u + w \Rightarrow v = w$) and sharpness ($u + v = 0 \Rightarrow u = v = 0$) pass from the T_α to T. Saturation: if $g = u - v \in T^{\mathrm{gp}}$ with $u, v \in T$ and $ng \in T$, say $nu = nv + w$ with $w \in T$, lift u, v, w and this equality to a stage T_α; there $n(u - v) = w \in T_\alpha$, so $u - v \in T_\alpha$ by saturation, and its image in T is g. Thus T lies in **Mon**, and it is then the colimit there. (Integrality and saturation are also covered by [4, Prop. I.1.3.6].) □

Two formal consequences of (3), both proved by comparing the functors $\mathrm{Hom}(-, T)$, are used repeatedly: for subsets $S, S' \subset G$ and a homomorphism $u : R \to R'$,

$$R\langle S\rangle \amalg_R R\langle S'\rangle \simeq R\langle S \cup S'\rangle, \tag{4}$$

$$R' \amalg_R R\langle S\rangle \simeq R'\langle u^{\mathrm{gp}}(S)\rangle. \tag{5}$$

Indeed, a map out of the left side of (4) to T is a map $a : R \to T$ with $a^{\mathrm{gp}}(S) \subset T$ and $a^{\mathrm{gp}}(S') \subset T$, and a map out of the left side of (5) is a map $b : R' \to T$ with $(bu)^{\mathrm{gp}}(S) = b^{\mathrm{gp}}(u^{\mathrm{gp}}(S)) \subset T$. In particular $R\langle S\rangle\langle u^{\mathrm{gp}}(S')\rangle \simeq R\langle S \cup S'\rangle$ for $u : R \to R\langle S\rangle$.

The remaining two lemmas of this section are the algebraic heart of the argument: the first produces valuative monoids, the second says that adjoining a sign is harmless from the point of view of which elements become zero.

Lemma 4.4 (Extension of an order)**.** *Every monoid R embeds into a valuative monoid W with $W^{\mathrm{gp}} = R^{\mathrm{gp}}$.*

This is [2, Lemma 2.1.2.11]; we include the short proof.

Proof. Let $G = R^{\mathrm{gp}}$. Saturated sharp submonoids of G containing R form a poset under inclusion in which every chain has an upper bound, its union (saturation and sharpness are conditions on finitely many elements at a time). By Zorn's lemma choose a maximal one, C. We claim $C \cup (-C) = G$, so that $W = C$ is valuative with $W^{\mathrm{gp}} = G$.

Suppose that neither x nor $-x$ belongs to C. We show that $C' = (C + \mathbb{N}x)^{\mathrm{sat}}$ is still sharp; since C' is saturated, contains R, and contains $x \notin C$, this contradicts the maximality of C. Let $u, -u \in C'$. Then there are $m, n > 0$, $a, b \in C$ and $i, j \geq 0$ such that

$$mu = a + ix, \qquad -nu = b + jx.$$

Multiplying the first equation by n, the second by m, and adding,

$$0 = na + mb + (ni + mj)x.$$

If $ni+mj > 0$, then $-(ni+mj)x = na+mb \in C$, and saturation of C gives $-x \in C$, a contradiction. Otherwise $i = j = 0$, so mu and $-nu$ lie in C; then mnu and $-mnu$ lie in C, sharpness of C gives $mnu = 0$, and torsion-freeness of G gives $u = 0$. □

Call a homomorphism of monoids $f : Q \to Q'$ *zero-reflecting* if $f^{-1}(0) = \{0\}$. (Between sharp monoids this is the condition that f reflects units, called *local* in [4, p. 15].)

Lemma 4.5. *Let $x \in R^{\mathrm{gp}}$, and set*

$$R_+ = R\langle x\rangle, \qquad R_- = R\langle -x\rangle, \qquad R_0 = R\langle x, -x\rangle.$$

1. *If neither x nor $-x$ lies in R, the three maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting.*

2. *If $x \in R$, then $R_+ = R$, and $R_0 = R_-$ with $R_- \to R_0$ the identity and $R_+ \to R_0$ the map $R \to R_-$. Symmetrically, if $-x \in R$, then $R_- = R$ and $R_0 = R_+$.*

Proof. (2) If $x \in R$, then $R + \mathbb{N}x = R$, which is saturated and sharp, so $R_+ = R$; and $R\langle x, -x\rangle = R\langle -x\rangle$ by (2), since $\mathbb{N}x \subset R$. The description of the maps follows from the uniqueness in Lemma 4.2. The case $-x \in R$ is obtained by replacing x with $-x$.

(1) Since $R \to R_0$ factors through $R \to R_+$ and through $R \to R_-$ (Lemma 4.2), it suffices to treat R_0. Before sharpening, R_0 is $C = (R + \mathbb{Z}x)^{\mathrm{sat}}$. If $r \in R$ maps to zero in $C^\sharp$, then $-r \in C$, so

$$-nr = a + kx \qquad (n > 0,\ a \in R,\ k \in \mathbb{Z}).$$

If $k > 0$, then $-kx = a + nr \in R$, and saturation gives $-x \in R$; if $k < 0$, then $(-k)x = a + nr \in R$ gives $x \in R$. Both are excluded. If $k = 0$, then nr and $-nr$ lie in R, so $nr = 0$ by sharpness and $r = 0$ by torsion-freeness. □

5 A topology finer than the birhizomic topology

Definition 5.1. For an object R of A_P and $x \in R^{\mathrm{gp}}$, the *sign cover* of R at x is the family

$$\{\, R \longrightarrow R\langle x\rangle, \quad R \longrightarrow R\langle -x\rangle \,\} \qquad \text{(written in the monoid direction).} \tag{6}$$

Let τ be the Grothendieck topology on A_P^{op} generated by the sign covers, i.e. the smallest topology for which the sieve generated by every sign cover is covering.

By (5), the base change of the sign cover at x along $u : R \to R'$ (a fibre product in A_P^{op}) is the sign cover of R' at $u^{\mathrm{gp}}(x)$: *sign covers are stable under base change*. This gives τ an explicit description. A *tree family* over R is the family of leaves $\{R \to T_\ell\}_{\ell\in\Lambda}$ of a finite binary tree with root R in which the two children of each internal vertex M form a sign cover of M ($R \to T_\ell$ being the composite along the path from the root); its *depth* is the depth of the tree, the trivial tree (depth 0) giving $\{\mathrm{id}_R\}$. Base change along $u : R \to R'$ takes a tree family over R to a tree family over R' of the same depth (apply (5) vertex by vertex), and replacing each leaf of a tree family by a tree family over that leaf (grafting) gives again a tree family. Consequently the sieves containing a tree family form a Grothendieck topology (the three axioms—maximal sieve, stability under base change, local character, i.e. transitivity—follow from the trivial tree, base change and grafting respectively), in which sign covers are covering; conversely, in any topology in which sign covers are covering, every tree family generates a covering sieve, by induction on the depth using the local character axiom. Hence

$$\text{a sieve on } R \text{ is } \tau\text{-covering if and only if it contains a tree family over } R. \tag{7}$$

For τ, or any other topology J on A_P^{op}, we say that a family over R is a *J-cover* if the sieve it generates is J-covering. Thus a family over R is a τ-cover as soon as it is *refined* by a tree family, meaning that every leaf T_ℓ receives a map under R from some member of the family.

Members of tree families are epimorphisms, being composites of the epimorphisms $M \to M\langle \pm x\rangle$ (Lemma 4.2). Recall that a presheaf $\mathcal{F}$ on A_P^{op} satisfies the *sheaf condition* for a family $\{R \to T_\ell\}_{\ell\in\Lambda}$ if $\mathcal{F}(R) \to \prod_\ell \mathcal{F}(T_\ell)$ is injective with image the *compatible* families (s_ℓ), those for which s_ℓ and s_m have the same image in $\mathcal{F}(T_\ell \amalg_R T_m)$ for all ℓ, m (fibre products in A_P^{op} being pushouts in A_P); when $R \to T_\ell$ is an epimorphism, $T_\ell \amalg_R T_\ell = T_\ell$ and the condition for $\ell = m$ is vacuous. For a sign cover, compatibility of (s_+, s_-) means that s_+ and s_- have the same image in $\mathcal{F}(R\langle x\rangle \amalg_R R\langle -x\rangle) = \mathcal{F}(R\langle x, -x\rangle)$ (4).

Lemma 5.2 (Sheaf criterion)**.** *A presheaf $\mathcal{F}$ of sets or of abelian groups on A_P^{op} is a τ-sheaf as soon as it satisfies the sheaf condition for every sign cover* (6)*.*

Proof.

> **Reviewer 3**
>
> Exp: I would say this does not need so long of a proof, as this is completely standard.

Step A: $\mathcal{F}$ satisfies the sheaf condition for every tree family, over every object. We induct on the depth d; depth 0 is trivial and depth 1 is the hypothesis. Let $\{R \to T_\ell\}$ have depth $d+1 \geq 2$, let $R_\pm$ be the children of the root and $\Lambda = \Lambda_+ \sqcup \Lambda_-$ the leaves of the two subtrees, which are tree families over $R_\pm$ of depth $\leq d$. Let (s_ℓ) be compatible. For $\ell, m \in \Lambda_+$, restricting along the canonical map $T_\ell \amalg_R T_m \to T_\ell \amalg_{R_+} T_m$ shows that $(s_\ell)_{\ell\in\Lambda_+}$ is compatible over R_+, so by induction there is a unique $s_+ \in \mathcal{F}(R_+)$ restricting to s_ℓ for all $\ell \in \Lambda_+$; similarly s_-. We claim that s_+ and

s_- have the same image in $\mathcal{F}(R_0)$, $R_0 = R_+ \amalg_R R_-$. The base change of the R_+-subtree along $R_+ \to R_0$ is a tree family over R_0 of depth $\le d$ with leaves $T_\ell \amalg_{R_+} R_0 = T_\ell \amalg_R R_-$ ($\ell \in \Lambda_+$), so by the injectivity part of the induction hypothesis it suffices to compare the restrictions of s_+ and s_- to each $T_\ell \amalg_R R_-$; these are the restrictions of s_ℓ and of s_-. In turn, the base change of the R_--subtree along $R_- \to T_\ell \amalg_R R_-$ is a tree family of depth $\le d$ with leaves $T_\ell \amalg_R T_m$ ($m \in \Lambda_-$), on which s_ℓ and s_- restrict to the images of s_ℓ and s_m, which agree by compatibility; injectivity once more proves the claim. By the sheaf condition for the sign cover $\{R \to R_\pm\}$ there is a unique $s \in \mathcal{F}(R)$ restricting to $s_\pm$, hence to all s_ℓ. If $s' \in \mathcal{F}(R)$ also restricts to all s_ℓ, then $s'|_{R_\pm} = s_\pm$ by the uniqueness in the induction hypothesis, so $s' = s$.

Step B: $\mathcal{F}$ is a τ-sheaf (this is the usual comparison between sheaves for a site and for its associated topology, cf. [5, Tag 00ZC]; we include the argument since tree families, not being closed under isomorphism, do not form a site in the sense of that tag). Let Σ be a τ-covering sieve on R; by (7) it contains a tree family $\{R \to T_\ell\}$. A compatible family on Σ is a collection of sections $s_V \in \mathcal{F}(V)$, one for each arrow $(R \to V) \in \Sigma$ (sections are indexed by arrows; we write s_V when the arrow is clear), with $s_{V'} = s_V|_{V'}$ for every $V \to V'$ under R, and we must show that $s \mapsto (s|_V)_V$ is a bijection from $\mathcal{F}(R)$ onto such collections. Injectivity follows from Step A for $\{R \to T_\ell\}$. Given (s_V), the sections s_{T_ℓ} are compatible (both s_{T_ℓ} and s_{T_m} restrict to $s_{T_\ell \amalg_R T_m}$), so Step A gives $s \in \mathcal{F}(R)$ with $s|_{T_\ell} = s_{T_\ell}$ for all ℓ. For $(R \to V) \in \Sigma$, the sections $s|_V$ and s_V both restrict to $s_{V \amalg_R T_\ell}$ on each leaf of the base-changed tree family $\{V \to V \amalg_R T_\ell\}$, hence are equal by Step A. □

Our goal in this section is to show that τ is finer than J_{b}. The next proposition concerns finitely presented families that "see" all valuative monoids; by (B2), it applies to birhizomic covers. It contains the compactness argument needed for possibly infinitely generated R.

Proposition 5.3. *Let R be an object of A_P and let $\{R \to Q_i\}_{i \in I}$ be a family of maps of finite presentation such that every homomorphism $R \to V$ to a valuative monoid V factors through some $R \to Q_i$. Then the family is refined by a tree family; in particular it is a τ-cover.*

Proof. Step 1: a compact space of orders. Let $G = R^{\mathrm{gp}}$, and let $\mathcal{V}(R)$ be the set of saturated submonoids $D \subset G$ with

$$R \subset D, \qquad D \cup (-D) = G.$$

Such a D is a "total precone": it induces a total preorder on G, but it need not be sharp, and it is essential to allow this: the precones D_ϵ produced in Step 5 below contain the kernel of a surjection $G \to T_\epsilon^{\mathrm{gp}}$ and are not sharp in general. Identify D with its characteristic function and regard $\mathcal{V}(R)$ as a subset of the product $\{0,1\}^G$, compact by Tychonoff's theorem. It is closed: each of the conditions $r \in D$ (for $r \in R$), $0 \in D$, "$g, h \in D \Rightarrow g + h \in D$", "$ng \in D \Rightarrow g \in D$" and "$g \in D$ or $-g \in D$" involves finitely many coordinates, and $\mathcal{V}(R)$ is the intersection of these closed conditions. Hence $\mathcal{V}(R)$ is compact. (The same device, embedding a set of valuation-like objects into a Boolean product, appears in [1, Lemma 5.3.6].) For finite $S \subset G$, the subset

$$U_S = \{D \in \mathcal{V}(R) : S \subset D\}$$

is open and closed, being defined by finitely many coordinates.

Step 2: the valuative monoid of a precone as a filtered colimit. For $D \in \mathcal{V}(R)$, write $H_D = D \cap (-D)$ and $V_D = D/H_D = D^\sharp$. Then V_D is a monoid (integral, saturated, sharp, cf. §3.1),

valuative because $D \cup (-D) = G$, and a monoid under R via $R \subset D \to V_D$. We claim that, in the category of monoids under R,

$$V_D = \varinjlim_{\substack{S \subset D \\ S \text{ finite}}} R\langle S \rangle, \tag{8}$$

the colimit being over the directed set of finite subsets of D, with transition maps $R\langle S\rangle \to R\langle S'\rangle$ for $S \subset S'$ given by Lemma 4.2. Indeed, by (3) a compatible family of maps $R\langle S\rangle \to T$ under $a : R \to T$ is the same as the condition $a^{\mathrm{gp}}(D) \subset T$. Given this condition, $a^{\mathrm{gp}}|_D : D \to T$ kills H_D (as T is sharp) and so factors uniquely through V_D; conversely a map $V_D \to T$ under R composed with $D \to V_D$ shows $a^{\mathrm{gp}}(D) \subset T$. So V_D represents the functor $\varprojlim_S \mathrm{Hom}_R(R\langle S\rangle, -)$, which is the claim.

Step 3: finite presentation yields finite stages. Fix $D \in \mathcal{V}(R)$. By hypothesis the valuative map $R \to V_D$ factors as $R \to Q_i \to V_D$ for some i. Since (8) is a filtered colimit of monoids under R and $\mathrm{Hom}_R(Q_i, -)$ commutes with such colimits (finite presentation; see the discussion following Lemma 4.2), the map $Q_i \to V_D$ factors as

$$Q_i \longrightarrow R\langle S_D\rangle \longrightarrow V_D$$

under R, for some finite $S_D \subset D$. The clopen sets U_{S_D}, $D \in \mathcal{V}(R)$, cover $\mathcal{V}(R)$, since $D \in U_{S_D}$. By compactness choose $D_1, \ldots, D_m$ with $\mathcal{V}(R) = U_{S_1} \cup \cdots \cup U_{S_m}$, where $S_j = S_{D_j}$, and retain the maps

$$Q_{i_j} \longrightarrow R\langle S_j\rangle \qquad (j = 1, \ldots, m) \tag{9}$$

under R. Put $S = S_1 \cup \cdots \cup S_m$, a finite subset of G.

Step 4: the sign tree. Enumerate $S = \{s_1, \ldots, s_k\}$ and form the binary tree of depth k whose vertex at level ℓ with sign history $\epsilon(s_1), \ldots, \epsilon(s_\ell)$ is $R\langle \epsilon(s_1)s_1, \ldots, \epsilon(s_\ell)s_\ell\rangle$; by (5) and (4), the sign cover of this vertex at the image of $s_{\ell+1}$ has as members the two vertices at the next level. The leaves are

$$T_\epsilon = R\langle\, \epsilon(s)s : s \in S \,\rangle, \qquad \epsilon : S \longrightarrow \{1, -1\}, \tag{10}$$

so $\{R \to T_\epsilon\}_\epsilon$ is a tree family. It remains to show that each $R \to T_\epsilon$ factors through some $R\langle S_j\rangle$; composing with (9) then shows that the tree family refines $\{R \to Q_i\}$.

Step 5: each leaf factors through some $R\langle S_j\rangle$. The idea is to attach to each leaf T_ϵ a precone $D_\epsilon \in \mathcal{V}(R)$ extending the inequalities $\epsilon(s)s \geq 0$ that define T_ϵ; the finite subcover gives $S_j \subset D_\epsilon$ for some j, and we then check that the elements of S_j are nonnegative already in T_ϵ. Fix ϵ. By Lemma 4.4, embed T_ϵ into a valuative monoid W_ϵ with $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$. Let $D_\epsilon \subset G$ be the inverse image of W_ϵ under the group homomorphism $G \to T_\epsilon^{\mathrm{gp}} = W_\epsilon^{\mathrm{gp}}$. Then $D_\epsilon \in \mathcal{V}(R)$: it contains R (which maps into $T_\epsilon \subset W_\epsilon$), it is saturated (as the inverse image of a saturated monoid under a group homomorphism), and $D_\epsilon \cup (-D_\epsilon) = G$ because $W_\epsilon \cup (-W_\epsilon) = W_\epsilon^{\mathrm{gp}}$. Choose j with $D_\epsilon \in U_{S_j}$, i.e. $S_j \subset D_\epsilon$.

We check that every $s \in S_j$ becomes nonnegative in T_ϵ. Let $\bar{s} \in T_\epsilon^{\mathrm{gp}}$ be the image of s. If $\epsilon(s) = 1$, then $\bar{s} \in T_\epsilon$ by construction of T_ϵ. If $\epsilon(s) = -1$, then $-\bar{s} \in T_\epsilon \subset W_\epsilon$, while $s \in D_\epsilon$ says $\bar{s} \in W_\epsilon$; sharpness of W_ϵ gives $\bar{s} = 0$ in $W_\epsilon^{\mathrm{gp}} = T_\epsilon^{\mathrm{gp}}$, so again $\bar{s} \in T_\epsilon$. (Here it matters that W_ϵ has the *same* group completion as T_ϵ: a map to an arbitrary valuative monoid would only give $\bar{s} \mapsto 0$ there.) By Lemma 4.2 there is a map $R\langle S_j\rangle \to T_\epsilon$ under R, as required. □

Corollary 5.4. *Every birhizomic covering sieve on A_P^{op} is τ-covering; that is, J_{b} is coarser than τ, $J_{\mathrm{b}} \subset \tau$. Consequently every τ-sheaf is a J_{b}-sheaf.*

Proof. Let $\{R \to Q_i\}_{i\in I}$ be a birhizomic cover and let $I' \subset I$ be the set of indices for which $R \to Q_i$ is of finite presentation. By (B1) and the definition of a birhizomic cover, every homomorphism from R to a valuative monoid factors through some Q_i with $i \in I'$. Proposition 5.3 applied to $\{R \to Q_i\}_{i\in I'}$ shows that the sieve generated by this subfamily, hence also the sieve generated by the whole family, is τ-covering. Every J_{b}-covering sieve contains a birhizomic cover and is therefore τ-covering. The last assertion follows, since a τ-sheaf satisfies the sheaf condition for every τ-covering sieve, in particular for every J_{b}-covering one. □

6 The Kan extension and its Mayer–Vietoris sequences

Fix a presheaf of abelian groups F on B_P^{op}, i.e. a functor $F : B_P \to \mathrm{Ab}$, and let $\mathcal{G} = \mathrm{Lan}_j F$ be its left Kan extension (1) to A_P, a presheaf of abelian groups on A_P^{op}:

$$\mathcal{G}(P \to Q) = \varinjlim_{(L_a \to Q) \in B_P/Q} F(L_a).$$

We first make the indexing category explicit. By the universal property of $L_a = P[-a]^\sharp$, a map $L_a \to Q$ under P exists if and only if the image of a in Q is zero, and it is then unique. In particular B_P is a *thin* category (a preorder): there is at most one map $L_a \to L_b$, and it exists if and only if $a \mapsto 0$ in L_b. So B_P/Q is the preorder of elements $a \in P$ that die in Q (elements giving the same object of B_P being identified), with $a \le b$ when a dies in L_b. It is nonempty, since it contains $L_0 = P$, and directed: if a and b die in Q, so does $a + b$, and $a, b \mapsto 0$ in L_{a+b} because the images of a and b in the sharp monoid L_{a+b} sum to zero. Thus (1) is a filtered colimit, and since filtered colimits of abelian groups are computed on underlying sets, the Kan extension of F in abelian presheaves is the same as in presheaves of sets, as announced in the introduction.

Two consequences will be used.

Lemma 6.1. *1. For $Q = L_a$ in B_P, the comma category B_P/L_a has the terminal object (L_a, id), so the unit of the Kan extension is an isomorphism*

$$F(L_a) \xrightarrow{\sim} \mathcal{G}(L_a). \tag{11}$$

2. If $Q \to Q'$ is a zero-reflecting map under P, then an element $a \in P$ dies in Q if and only if it dies in Q', so the comma categories B_P/Q and B_P/Q' coincide, and the restriction map

$$\mathcal{G}(Q) \xrightarrow{\sim} \mathcal{G}(Q') \tag{12}$$

is an isomorphism (it is the identity of the common colimit).

Proof. (1) Every object $(L_b, L_b \to L_a)$ of B_P/L_a has the given map as its unique map to (L_a, id), by thinness; a colimit over a category with a terminal object is the value at that object, and the unit is the corresponding structure map. (2) An element killed in Q is killed in Q'; conversely, if $a \mapsto 0$ in Q', then the image of a in Q is killed by the zero-reflecting map $Q \to Q'$, so it is 0. The restriction map $\mathcal{G}(Q) \to \mathcal{G}(Q')$ is induced by the identity of the diagrams, hence is the identity. □

In words, $\mathcal{G}(Q)$ depends only on the set $\{a \in P : a \text{ dies in } Q\}$ of elements of P killed by Q (a *face* of P: a submonoid containing a and b whenever it contains $a + b$), and $\mathcal{G}$ is "locally constant" along zero-reflecting maps. Combined with Lemma 4.5, which says that sign covers consist of zero-reflecting maps except in the degenerate case, this gives the sheaf property and more.

Lemma 6.2. *For every sign cover* (6)*, with $R_\pm$ and R_0 as in Lemma 4.5, the sequence*

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{G}(R_+) \oplus \mathcal{G}(R_-) \longrightarrow \mathcal{G}(R_0) \longrightarrow 0, \tag{13}$$

whose first map is the pair of restrictions and whose second map is the difference of the two restrictions to R_0, is exact, and in fact split. Moreover $\mathcal{G}$ is a τ-sheaf, hence (Corollary 5.4) a J_{b}-sheaf.

Proof. Suppose first that neither x nor $-x$ lies in R. By Lemma 4.5(1) the maps $R \to R_+$, $R \to R_-$ and $R \to R_0$ are zero-reflecting, so by (12) the three restriction maps out of $\mathcal{G}(R)$ are isomorphisms; since $\mathcal{G}(R) \to \mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ composes to the isomorphism $\mathcal{G}(R) \to \mathcal{G}(R_0)$, the maps $\mathcal{G}(R_\pm) \to \mathcal{G}(R_0)$ are isomorphisms too. Identifying all four groups with $A = \mathcal{G}(R)$ through these isomorphisms, every restriction map becomes the identity of A and the sequence becomes

$$0 \longrightarrow A \xrightarrow{a \mapsto (a,a)} A \oplus A \xrightarrow{(a,b) \mapsto a-b} A \longrightarrow 0,$$

which is exact, and split by $a \mapsto (a, 0)$.

If $x \in R$, then by Lemma 4.5(2) $R_+ = R$, $R_0 = R_-$, the map $R_- \to R_0$ is the identity and $R_+ \to R_0$ is $R \to R_-$. Writing $A = \mathcal{G}(R)$, $B = \mathcal{G}(R_-)$ and $\rho : A \to B$ for the restriction, the sequence reads

$$0 \longrightarrow A \xrightarrow{a \mapsto (a, \rho(a))} A \oplus B \xrightarrow{(a,b) \mapsto \rho(a) - b} B \longrightarrow 0,$$

which is exact, split by $(a, b) \mapsto a$, for any homomorphism ρ. The case $-x \in R$ is the same with the roles of R_+ and R_- exchanged.

For the sheaf property, by Lemma 5.2 it suffices to check the sheaf condition for sign covers, and as noted before that lemma this condition says that $\mathcal{G}(R)$ is the equalizer of the two restriction maps $\mathcal{G}(R_+) \times \mathcal{G}(R_-) \rightrightarrows \mathcal{G}(R_0)$, which is exactness of (13) at the first two terms. □

7 Acyclicity, and passage to the coarser topology

For a Grothendieck topology J on A_P^{op}, an object R and an abelian J-sheaf $\mathcal{E}$, we write $H^n_J(R, \mathcal{E})$ for the nth right derived functor of $\mathcal{E} \mapsto \mathcal{E}(R)$ on the abelian category $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, J)$ of abelian J-sheaves, computed by injective resolutions. We shall use two standard facts. First, the Mayer–Vietoris sequence [5, Tag 0EVY] for a cover $\{R_\pm \to R\}$ in A_P^{op} by two monomorphisms ($R \to R_\pm$ being epimorphisms in A_P) with fibre product R_0; we spell out the version we need in the proof below. Second, *locality of cohomology* [5, Tag 03P1]: for $n > 0$, every class $\xi \in H^n_J(R, \mathcal{E})$ vanishes on the members of some J-cover of R (in the sense of §5). (Explicitly: if $\mathcal{E} \to \mathcal{I}^\bullet$ is an injective resolution and $z \in \mathcal{I}^n(R)$ a cocycle representing ξ, the exactness of $\mathcal{I}^\bullet$ as a complex of *sheaves* in degree $n > 0$ says that z is locally a coboundary.)

Lemma 7.1. *Let $\mathcal{E}$ be an abelian τ-sheaf on A_P^{op} such that for every sign cover* (6)*, with $R_\pm$ and R_0 as in Lemma 4.5, the difference map*

$$\mathcal{E}(R_+) \oplus \mathcal{E}(R_-) \longrightarrow \mathcal{E}(R_0) \tag{14}$$

is surjective. Then $H^n_\tau(R, \mathcal{E}) = 0$ for all $n > 0$ and all objects R of A_P.

Proof. Mayer–Vietoris. For an object Y of A_P let h_Y be the presheaf represented by Y on A_P^{op}, so $h_Y(X) = \mathrm{Hom}_{A_P}(Y, X)$, and let $\mathbb{Z}_\tau[Y]$ be the τ-sheaf associated with the free abelian presheaf $\mathbb{Z}[h_Y]$; then $\mathrm{Hom}(\mathbb{Z}_\tau[Y], \mathcal{E}) = \mathcal{E}(Y)$ for every abelian τ-sheaf $\mathcal{E}$ [5, Tag 03AB], so that $H^n_\tau(Y, \mathcal{E}) = \mathrm{Ext}^n(\mathbb{Z}_\tau[Y], \mathcal{E})$, the Ext group being computed in $\mathrm{Ab}\,\mathrm{Sh}(A_P^{\mathrm{op}}, \tau)$ (apply $\mathrm{Hom}(\mathbb{Z}_\tau[Y], -)$ to an injective resolution of $\mathcal{E}$). Fix a sign cover of R at x. For every X, the universal property (3) identifies $h_{R_\pm}(X)$ with subsets of $h_R(X)$, with intersection $h_{R_0}(X)$ by (4). The elementary exact sequence $0 \to \mathbb{Z}[U \cap U'] \to \mathbb{Z}[U] \oplus \mathbb{Z}[U'] \to \mathbb{Z}[U \cup U'] \to 0$ of free abelian groups on two subsets of a set therefore gives an exact sequence of presheaves

$$0 \longrightarrow \mathbb{Z}[h_{R_0}] \longrightarrow \mathbb{Z}[h_{R_+}] \oplus \mathbb{Z}[h_{R_-}] \longrightarrow \mathbb{Z}[h_R]$$

whose cokernel is $\mathbb{Z}[h_R]/\mathbb{Z}[h_{R_+} \cup h_{R_-}]$. This cokernel has zero τ-sheafification: a section of h_R over X is a map $f : R \to X$, and on the two members of the sign cover of X at $f^{\mathrm{gp}}(x)$, which are $X \amalg_R R_\pm$ by (5), the restrictions of f lie in h_{R_+} and h_{R_-} respectively. Since sheafification is exact we obtain a short exact sequence of τ-sheaves

$$0 \longrightarrow \mathbb{Z}_\tau[R_0] \longrightarrow \mathbb{Z}_\tau[R_+] \oplus \mathbb{Z}_\tau[R_-] \longrightarrow \mathbb{Z}_\tau[R] \longrightarrow 0,$$

and applying $\mathrm{Ext}^\bullet(-, \mathcal{E})$ the long exact sequence

$$\begin{aligned} 0 \to \mathcal{E}(R) \to \mathcal{E}(R_+) \oplus \mathcal{E}(R_-) \to \mathcal{E}(R_0) \to H^1_\tau(R, \mathcal{E}) \to \cdots \\ \cdots \to H^{n-1}_\tau(R_0, \mathcal{E}) \to H^n_\tau(R, \mathcal{E}) \to H^n_\tau(R_+, \mathcal{E}) \oplus H^n_\tau(R_-, \mathcal{E}) \to H^n_\tau(R_0, \mathcal{E}) \to \cdots \end{aligned} \tag{15}$$

This is the Mayer–Vietoris sequence of [5, Tag 0EVY] for the present situation.

Injectivity on the members of a sign cover. We prove the lemma by induction on $n \geq 1$, simultaneously for all objects R. Consider the statement $(*)_n$ that, for every sign cover of every object R, the map

$$H^n_\tau(R, \mathcal{E}) \longrightarrow H^n_\tau(R_+, \mathcal{E}) \oplus H^n_\tau(R_-, \mathcal{E})$$

is injective. For $n = 1$, $(*)_1$ follows from (15) and the assumed surjectivity of (14), which makes the connecting map $\mathcal{E}(R_0) \to H^1_\tau(R, \mathcal{E})$ zero. For $n > 1$, if $H^{n-1}_\tau(-, \mathcal{E}) = 0$ on all objects, then in particular $H^{n-1}_\tau(R_0, \mathcal{E}) = 0$ and (15) gives $(*)_n$.

From the leaves to the root. Having established $(*)_n$ (unconditionally for $n = 1$, and under the induction hypothesis for $n > 1$), let $\xi \in H^n_\tau(R, \mathcal{E})$. By locality of cohomology, ξ vanishes on the members of some τ-cover of R. By (7), the sieve generated by this family contains a tree family $\{R \to T_\ell\}$ (each T_ℓ receives a map under R from a member on which ξ vanishes), so $\xi|_{T_\ell} = 0$ for every leaf. Now argue upward through the tree: if M is an internal vertex with children M_+, M_- forming a sign cover of M, and $\xi|_{M_+} = 0 = \xi|_{M_-}$, then $\xi|_M = 0$ by $(*)_n$ applied to the sign cover of M. Induction on the height of a vertex gives $\xi|_R = \xi = 0$. Hence $H^n_\tau(R, \mathcal{E}) = 0$ for all R, which completes the induction on n. □

Corollary 7.2. *Let $\mathcal{G} = \mathrm{Lan}_j F$ be the Kan extension of §6. For every object R of A_P and every $n > 0$, $H^n_\tau(R, \mathcal{G}) = 0$.*

Proof. By Lemma 6.2, $\mathcal{G}$ is a τ-sheaf whose maps (14) are surjective. Apply Lemma 7.1. □

We now transfer this vanishing to the coarser topology J_{b}. Some care is needed, since the inclusion of τ-sheaves into J_{b}-sheaves is not exact in general (cokernels involve different sheafifications); what does transfer is injectivity of objects, and Corollary 7.2 makes a τ-injective resolution exact already as a complex of presheaves.

Proposition 7.3. *For $\mathcal{G} = \operatorname{Lan}_j F$, every object R of A_P and every $n > 0$, $H^n_{J_\mathrm{b}}(R, \mathcal{G}) = 0$.*

Proof. Since $J_\mathrm{b} \subset \tau$ (Corollary 5.4), the identity of A_P^{op} is a morphism of sites $(A_P^{\mathrm{op}}, \tau) \to (A_P^{\mathrm{op}}, J_\mathrm{b})$ whose direct image is the inclusion

$$\iota : \operatorname{Ab}\operatorname{Sh}(A_P^{\mathrm{op}}, \tau) \longrightarrow \operatorname{Ab}\operatorname{Sh}(A_P^{\mathrm{op}}, J_\mathrm{b})$$

and whose inverse image is τ-sheafification, an exact functor [5, Tag 0EWK]. A right adjoint whose left adjoint is exact preserves injectives [5, Tag 015Z]; hence ι carries injective τ-sheaves to injective J_b-sheaves.

Choose an injective resolution $\mathcal{G} \to \mathcal{I}^\bullet$ in $\operatorname{Ab}\operatorname{Sh}(A_P^{\mathrm{op}}, \tau)$. By Corollary 7.2, for every object R the augmented complex of abelian groups

$$0 \longrightarrow \mathcal{G}(R) \longrightarrow \mathcal{I}^0(R) \longrightarrow \mathcal{I}^1(R) \longrightarrow \cdots \tag{16}$$

is exact. So $\mathcal{G} \to \mathcal{I}^\bullet$ is exact as a complex of presheaves, hence exact as a complex of J_b-sheaves (kernels of maps of sheaves are computed in presheaves, and a presheaf image that is already a sheaf equals the sheaf image). Its terms $\iota\mathcal{I}^k$ are injective J_b-sheaves by the preceding paragraph. Thus $\mathcal{G} \to \iota\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{G}$ in $\operatorname{Ab}\operatorname{Sh}(A_P^{\mathrm{op}}, J_\mathrm{b})$, and $H^n_{J_\mathrm{b}}(R, \mathcal{G})$ is the nth cohomology of the same complex (16), which vanishes for $n > 0$. □

8 Proof of Theorem 3.1

Recall from §3.3 that $\pi^{-1}F$ is the J_b-sheafification of the left Kan extension $\mathcal{G} = \operatorname{Lan}_j F$, and that π_* is restriction along $j : B_P \hookrightarrow A_P$.

$\pi^{-1}F = \mathcal{G}$. By Lemma 6.2, $\mathcal{G}$ is already a J_b-sheaf, so no sheafification is needed and $\pi^{-1}F = \mathcal{G}$.

The unit in degree zero. The unit $F \to \pi_*\pi^{-1}F = \mathcal{G} \circ j$ evaluated at L_a is the Kan extension unit $F(L_a) \to \mathcal{G}(L_a)$, an isomorphism by (11).

Vanishing of $R^n\pi_*$ for $n > 0$. Let $\mathcal{G} \to \mathcal{I}^\bullet$ be an injective resolution in $\operatorname{Ab}\operatorname{Sh}(A_P^{\mathrm{op}}, J_\mathrm{b})$. By definition $\mathrm{R}\pi_*\mathcal{G}$ is represented by the complex of presheaves $\pi_*\mathcal{I}^\bullet = \mathcal{I}^\bullet \circ j$ on B_P^{op}. Since B_P^{op} carries the trivial topology, evaluation at any object L_a is an exact functor on $\operatorname{Ab}\operatorname{Sh}(B_P^{\mathrm{op}}) = \operatorname{Ab}\operatorname{PSh}(B_P^{\mathrm{op}})$, so

$$(R^n\pi_*\mathcal{G})(L_a) = H^n\big(\mathcal{I}^\bullet(L_a)\big) = H^n_{J_\mathrm{b}}(L_a, \mathcal{G}) = 0 \qquad (n > 0)$$

by Proposition 7.3. As the L_a are all the objects of B_P, the cohomology presheaves $R^n\pi_*\mathcal{G}$ vanish for $n > 0$, and $R^0\pi_*\mathcal{G} = \pi_*\mathcal{G} \cong F$ by the previous paragraph. Hence $F \to \mathrm{R}\pi_*\pi^{-1}F$ is a quasi-isomorphism. This proves Theorem 3.1, including the three "more precise" assertions. □

Remark 8.1 (Where the hypotheses are used). The infinitesimal lifting condition in the definition of a birhizomic cover is never analysed: Proposition 5.3 shows that *any* family of finitely presented maps through which every valuative map factors is a τ-cover, and this is all that is used. Finite presentation enters exactly once, in Step 3 of that proof, to factor $Q_i \to V_D$ through a finite stage of the colimit (8); the colimit identity itself needs no finiteness. Finite generation of P is never used, which is why the compactness argument on $\mathcal{V}(R)$ is needed in place of a dimension count. Finally, the argument of Proposition 7.3 shows that $\mathcal{G}$ is acyclic for every topology coarser than τ, the birhizomic topology being one instance.

Remark 8.2 (Two small examples). For $R = \mathbb{N}^2$ and $x = e_1 - e_2$, neither sign lies in R; the members of the sign cover are $\mathbb{N}^2\langle\pm(e_1 - e_2)\rangle$, the two charts of the blow-up of the origin, and $R_0 = \mathbb{N}^2\langle x, -x\rangle \simeq \mathbb{N}$ via $(a, b) \mapsto a + b$. None of these maps kills a nonzero element of $\mathbb{N}^2$, so $\mathcal{G}$ takes the same value on all four monoids, as in the first case of Lemma 6.2. For $R = \mathbb{N}$ and $x = 1$ the sign cover is $\{\mathbb{N} \to \mathbb{N},\ \mathbb{N} \to 0\}$ with $R_0 = 0$, and (13) is $0 \to \mathcal{G}(\mathbb{N}) \to \mathcal{G}(\mathbb{N}) \oplus \mathcal{G}(0) \to \mathcal{G}(0) \to 0$. Deeper in a sign tree a leaf may kill nonzero elements of P; this is harmless, as Lemma 4.5 is applied at each internal vertex.

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